

Local NMP User Guide

User Guide

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1 Introduction

Local Network Management Portal (Local NMP) is a locally hosted web portal designed to help users of Ericsson Private 5G with configuration, management, and monitoring of their wireless networks.

Local NMP supports and features the following:

- An intuitive user interface
- Overview of all network equipment
- Secure sign-in method with individual user credentials and Multi-Factor Authentication (MFA)
- Time and date fields based on the local time zone of the device used for accessing NMP

This document provides instructions on how to use Local NMP.

For more information on the applicable security measures regarding Local NMP, see the section *Local Management Portal* in the document *Security User Guide*.



2 Accessing Local NMP

Local Network Management Portal (Local NMP) can be accessed using the latest versions of Google Chrome or Mozilla Firefox, and the ESR version of Mozilla Firefox. Local NMP can only be accessed from within the VRF configured for local access in NMP. The IP addresses for accessing Local NMP are determined when configuring local access. The IP addresses are displayed and can be edited on the **Local Access** page of NMP.

Local NMP can be accessed through both network controllers, however, some of the statuses displayed can differ between the network controllers under certain circumstances, such as connectivity issues. The **Equipment page** displays a **Portal Host** label next to the network controller that is currently in use to access Local NMP.

A DNS or load balancer can also be used for connecting to Local NMP. In this case, it is recommended to configure the two network controllers as primary and secondary, so that the traffic is distributed to the primary network controller by default and only to the secondary network controller in case of a failover. Configuring a DNS or load balancer to distribute traffic in a random or round-robin way can cause functions such as email verification and SSO to fail.

Sign-In Methods

Note: All or some of this functionality requires site version 26.20 or later. For more information, see the document [Release Notes](#).

Accessing Local NMP works differently depending on the sign-in configuration of the site. The following methods are available for users to sign in to Local NMP:

- Email and password

Ericsson security applies, such as password aging, password rules, and MFA on or off.

For sites that use email and password with MFA enabled, an authenticator app is required for sign-in authentication. The authenticator app can be installed on a personal computing device, such as a smartphone or a laptop. Validated apps are Symantec VIP Access, Google Authenticator, and Microsoft Authenticator, but any app implementing the standard "RFC 6238 - TOTP: Time-Based One-Time Password Algorithm" should also work.

Note: If the QR-code scanning does not work, verify that the authenticator app has camera access.

- Using Single Sign-On (SSO) with an external Identity Manager (IDM)

The integration between Ericsson Private 5G and an external IDM relies on using the same user email addresses in both NMP and the external IDM.



Security depends on the used external IDM, provided by the enterprise or the partner. Ericsson security rules and procedures do not apply.

To log in with SSO, do not enter an email address and password on the general Local NMP sign-in page. Click the **Sign in with Single Sign-On (SSO)** button and enter the email address on the resulting page.

The sign-in configuration can be edited on the **Edit Site** page. For more information, see [Site Security](#) on page 9.

Break Glass Access

Note: All or some of this functionality requires site version 26.20 or later. For more information, see the document [Release Notes](#).

For sites configured with SSO only as the sign-in method, Local Admin users can sign in with email and password to recover the site in case of SSO failure.

For Local Admin users, the following procedure must be used in case a password is missing:

- Reset Password
- Reset Two-Factor Authentication
- Reset Recovery Codes

For more information, see [User Settings](#) on page 4.



3 User Menu

Note: All or some of this functionality requires site version 26.20 or later. For more information, see the document [Release Notes](#).

The user menu in Local NMP contains the following menu items:

- A link to the **User Settings** page
- A theme switch to toggle between light and dark mode
- A link to display the Ericsson Private 5G Ownership Notice text in the main window
- A **Sign out** button

3.1 User Settings

For information about the user role access rights for this page, see the section *Local Network Management Portal Access Rights* in the document [User Role Description](#).

Note: All or some of this functionality requires site version 26.20 or later. For more information, see the document [Release Notes](#).

The **User Settings** page allows users to view and edit their user account related information.

Steps

Under **User Information**, click on the pen icons to edit the information. After edits have been made, click ✓ to confirm or ✗ to discard edits. Changes are saved by clicking **Save** at the bottom of the page.

1. **Optional:** Edit the full name.
2. **Optional:** Edit the email address.

Note: If two-factor authentication is enabled, the secondary authentication methods must be reset for the authenticator app to use the new email address.

Under **Account Security**:

3. **Optional:** To reset the password, click **Reset** next to **Password**.

Result: When resetting the password, the user will automatically be relocated to the sign-in page for updating the password.



4. **Optional:** To reset the two-factor authentication device and the recovery codes, click **Reset** next to **Secondary Authentication Methods**.

Result: When resetting the secondary authentication methods, the user will automatically be relocated to the sign-in page for adding a new two-factor authentication device and receiving new recovery codes.



4 Edit Site

For information about the user role access rights for this page, see the section *Local Network Management Portal Access Rights* in the document *User Role Description*.

Note: All or some of this functionality requires site version 26.20 or later. For more information, see the document *Release Notes*.

The **Edit Site** page can be accessed from any page in Local NMP by clicking the action menu  to the right of the site name in the header of the page.

The following sections lists the information and settings that can be viewed or edited for active sites that are online. Click on the pen icons next to an element to edit the information. After edits have been made, click  to confirm edits or  to discard edits. Changes are saved by clicking **Save** at the bottom of the page.

Site Information Tab

Under the **Site Information** section:

- **Radio Network Mode** (View only)
- **Site Version** (View only)
- **Site ID** (View only)
- **Service Provider Name** (For more information, see [Service Provider Name](#) on page 8.)
- **Payload MTU** (For more information, see [Payload MTU](#) on page 8.)
- **Network Controller Redundancy Configuration** (View only)
- **Baseband Connectivity** (View only. Only for Click-to-Deploy sites)

Under the **License Information** section:

- **EUFT/Customer ID** (View only. For more information, see the section *Create Site* in the document *Installation Planning and Site Survey*.)
- **SWLT** (View only. For more information, see the section *Create Site* in the document *Installation Planning and Site Survey*.)

Site Security Tab

For more information, see [Site Security](#) on page 9.



Email Server Tab

- SMTP Server Address
- SMTP Server Port
- Authentication

Under the **Authentication Details** section if **Authentication** is enabled:

- Username
- Password

Under the **Encryption** section:

- SSL (Secure Sockets Layer)
- Start TLS

Log Streams Tab

The log stream starts after clicking **Save**. It is possible to stop and resume a stream from the **Log Streams** tab. Once a stream is stopped, it is possible to delete the destination.

Under the **EBM** section if **EBM Log Stream** is enabled:

- EBM Log Stream
- Destination Name
- IP Address
- Port
- Protocol (View only)
- Log Stream Status (View only)
- TLS Certificate
- Issued On/Expires On
- TLS Certificate Status

Under the **Security** section if **Security Log Stream** is enabled:

- Security Log Stream
- Destination Name
- IP Address



- Port
- Protocol (View only)
- Log Stream Status (View only)
- TLS Certificate
- Issued On/Expires On
- TLS Certificate Status

PLMN Tab (Only for Click-to-Deploy Sites)

Under the **EHPLMN Configuration** section:

- **EHPLMN MCC/MNC** (For more information, see [About Support for SIM Cards with External IMSIs](#) on page 10.)

4.1 Service Provider Name

The maximum character length for the service provider name is 16, and can be configured with the following ASCII characters:

- Numbers (0-9)
- Upper and lower case English letters (A-Z, a-z)
- Some special characters:
 - Space
 - Number sign (#)
 - Percent (%)
 - Forward slash (/)
 - Colon (:))
 - Question mark (?)

4.2 Payload MTU

Payload MTU is the maximum packet size a device can use. By default, Ericsson Private 5G networking uses a payload MTU size of 1500 bytes. To prevent misalignment, the VRFs on the enterprise LAN should be configured with an interface MTU size of at least 1500 bytes.



- Note:**
- Changing the payload MTU can result in fragmentation and packet loss, affecting performance.
 - Changes to payload MTU apply immediately, but devices do not use the new value until they disconnect and then reconnect.
 - Keep in mind that the payload MTU must be supported end-to-end, including the air interface, radio transport network, and enterprise LAN.

The payload MTU configuration available in Local NMP does not affect the MTU size of the radio-network equipment. For **Click-to-Deploy** sites, the radio-network equipment use a fixed setting of 1968 which works well with the supported payload MTU sizes available in Local NMP (1280 – 1920). The MTU size for the radio-network equipment should be at least 48 bytes more than the payload MTU.

Payload MTU for Decoupled Sites

For **Decoupled** sites, configuring the radio-network equipment to use an MTU of 1968 can be a good starting point, but the responsibility to configure this setting properly and avoid fragmentation belongs to the CSP or Ericsson customer unit handling the project.

4.3 Site Security

On the **Site Security** tab, click on the pen icons to edit the information. After edits have been made, click ✓ to confirm or ✗ to discard edits. Changes are saved by clicking **Save** at the bottom of the page.

Sign-In Security

The sign-in configuration allows for either email and password or SSO, or both to be enabled. Having both methods enabled is meant to be temporary, during a transition from using the email and password method to using SSO. Two successful SSO sign-ins by two unique users with both methods enabled are required before selecting only SSO. This is needed to verify that the integration between the external IDM and Local NMP works properly.

The SSO configuration can only be edited when the sign-in configuration is set to **Email/password + Single Sign-On**.

Note: When switching back from **Single Sign-On** to either **Email/password + Single Sign-On** or **Email/password**, users will have to request a new password before they can sign in with email and password. All currently signed in users are automatically signed out during the switch and must sign in again.

If **Email/password + Single Sign-On** or **Single Sign-On** is chosen as sign-in method, the following information and configuration items are visible:



- Redirect URI

The URI is used for identifying an Ericsson Private 5G site in an external IDM. By clicking the copy icon, the URI can be copied to the clipboard and pasted into the external IDM.

- Authentication Protocol

Ericsson Private 5G works with the Open ID Connect (OIDC) protocol. The external IDM must be compatible with this protocol.

- Generated in the external IDM:

- Client ID

The client ID works as a user name for the Ericsson Private 5G site in the external IDM. The client ID can be shared by multiple sites that use the same external IDM.

- Client Secret

The client secret is generated together with the client ID in the external IDM.

- Discovery Endpoint

The discovery endpoint is unique to the IDM instance and is used by Ericsson Private 5G to point to the external IDM.

4.4 About Support for SIM Cards with External IMSIs

SIM cards are provisioned specifically for each Ericsson Private 5G site. The IMSIs of the SIM cards are based on the PLMN ID (MCC and MNC) of the site.

SIM cards with IMSIs based on external PLMNs can also be used. The external PLMN ID is added during installation of Ericsson Private 5G, or anytime through NMP. To attach to the Ericsson Private 5G site, those SIM cards must have the PLMN ID of the Ericsson Private 5G site configured as an Equivalent Home PLMN (EHPLMN) ID. Up to five PLMNs and EHPLMNs can be configured on a network controller.



5 Monitoring

The **Monitoring** menu category includes the following menu items:

- Event Logs

5.1 Event Logs

For information about the user role access rights for this page, see the section *Local Network Management Portal Access Rights* in the document *User Role Description*.

The **Event Logs** page displays a history of system events and the time and date on which they occurred. For more information about the available events, see the document *Event Log and Email Notifications for NMP and Local NMP*.

Note: All or some of this functionality requires site version 26.20 or later. For more information, see the document *Release Notes*.



6 Network

The **Network** menu category includes the following menu items:

- Equipment
- VRFs
- Segments
- SIM Cards

6.1 Equipment

For information about the user role access rights for this page, see the section *Local Network Management Portal Access Rights* in the document *User Role Description*.

For sites using the **Click-to-Deploy** radio network mode, the **Equipment** page displays all connected, disconnected, or disabled Ericsson Private 5G equipment.

For sites using the **Decoupled** radio network mode, the **Equipment** page only displays information about the network controllers.

- Note:**
- For sites with site version 25.50 or earlier, routers are not visible in Local NMP. For sites with site version 25.80 or later, routers are visible without device information.
 - The hardware status for GNSS receivers will always be blank in the table on the **Equipment** page. To view the hardware status of a GNSS receiver, see [Equipment Details](#) on page 14.
 - A warning box will appear if radios, IRUs, or radio dots are disconnected.
 - The hardware status of a piece of equipment depends on different factors for different equipment. Hardware status down generally indicates that a piece of equipment has lost connectivity.
 - If the service status for a network controller is down, one or more services on the network controller are in a non-functional state, or the link is down between the radio network connection (Radio-0 interface) on the network controller and one or more CBRS basebands.



Edit Network Controllers

To edit the time synchronization settings for the site, click on the action menu  to the right of **Network Controllers** and select **Edit Network Controllers**. For more information, see [Time Synchronization Settings](#) on page 59.

Note: All or some of this functionality requires site version 26.20 or later. For more information, see the document [Release Notes](#).

Equipment Details

To access additional details for a piece of equipment, either click on the equipment name or click on the action menu  to the right of the equipment and select **View Details**. For more information, see [Equipment Details](#) on page 14.

Edit Equipment

To make configuration changes to a piece of equipment, click on the action menu  to the right of the equipment and select **Edit**. For more information, see [Edit Equipment](#) on page 16.

Note: All or some of this functionality requires site version 26.20 or later. For more information, see the document [Release Notes](#).

Disable or Enable Equipment

To disable or enable a piece of equipment, click on the action menu  to the right of the equipment and select **Disable** or **Enable**. Confirm by clicking **Disable** or **Enable**.

Delete Radio

To delete a radio, click on the action menu  to the right of the radio and select **Delete**. Confirm by clicking **Delete Radio**.

Note: This change is permanent. After deletion, any deleted radio equipment will stop radio transmissions.

There might be a brief delay between the deletion of the radio and its disappearance from the equipment list on the **Equipment** page.

When a radio has been deleted, the mobility configuration needs to be updated for changes to take effect. An information message appears on the equipment page when updates are needed.

Note: All or some of this functionality requires site version 26.20 or later. For more information, see the document [Release Notes](#).



Disable or Enable Cloud Connection

The cloud connection of each network controller can be disabled or enabled. Cloud connection is enabled by default and is reset to this default setting after a restart. Leaving cloud connection enabled is recommended since the cloud acts as a backup server and handles SLA calculations.

To disable or enable cloud connection for a network controller, click on the action menu  to the right of the network controller and select **Disable Cloud Connection** or **Enable Cloud Connection**. Confirm by clicking **Disable Cloud Connection** or **Enable Cloud Connection**.

Restart Equipment

If no other troubleshooting method is available, network controllers and basebands can be restarted.

Restarting a network controller can take up to 30 minutes and might cause a partial or complete network interruption. The network controller that is used for hosting Local NMP can also be restarted, in this case, Local NMP will disconnect after beginning the restart and will only reconnect once the restart has finished. The restarted network controller automatically attempts to connect to the cloud.

Restarting a baseband can take up to 5 minutes and interrupts traffic handled by the baseband. During a restart of a baseband, the radios connected to it are not displayed on the **Equipment** page.

To restart a network controller or baseband, click on the action menu  to the right of the network controller or baseband and select **Restart**. Confirm by typing **RESTART** and clicking **Restart Network Controller** or by clicking **Restart Baseband**.

Cells Tab

The **Cells** tab displays various information about all the cells configured on the radio network. To disable or enable a cell, click on the action menu  to the right of the cell and select **Disable** or **Enable**.

6.1.1

Equipment Details

For information about the user role access rights for this page, see the section *Local Network Management Portal Access Rights* in the document *User Role Description*.

The **Equipment Details** page contains cards that display general equipment information and information about the status of the selected piece of equipment, as well as an interactive console. This information can be useful for troubleshooting and when communicating with support. The available cards vary based on equipment type.



Download Logs

Note: All or some of this functionality requires site version 26.20 or later. For more information, see the document [Release Notes](#).

Logs can be prepared and downloaded for the network controllers. The logs are generated independently for each network controller. The exported logs can be used for troubleshooting.

Logs are prepared by clicking **Download Logs**. Select the log types to prepare and then click **Start Preparing Logs**. For more information about the log types, see [Network Controller Logs](#) on page 73.

Once the preparation is complete, the logs appear in the drop-down menu where they can be downloaded for seven days. Select the logs to download them to a local device.

Equipment Information

The **Equipment Information** card displays information relevant to the selected piece of equipment.

For basebands, the following synchronization statuses are applicable:

Synchronized	The baseband is receiving a time synchronization signal, either directly from a GNSS receiver or from a router connected to a GNSS receiver.
Disconnected	The baseband is not receiving a time synchronization signal.
Pending	The baseband is attempting to synchronize but have not achieved synchronization.

Because of a known limitation, sites that use Router 6675 can not display equipment information for GNSS receivers.

Some radio equipment (IRUs, radio dots, and radios) that was sold with Ericsson Private 5G 1.1 is no longer offered and has been replaced with equivalent radio equipment. The equivalent radio equipment has the exact same hardware specifications, software functionality, and life cycle management as the replaced radio equipment. The replaced radio equipment is still supported, but in NMP and in some documents, it is represented by the new equivalent radio equipment.

For more information, see [Replaced IRUs, Radio Dots, and Radios](#) on page 60.

Equipment Status

The **Equipment Status** card displays status information about the selected piece of equipment.



Port Information

Note: All or some of this functionality requires site version 26.20 or later. For more information, see the document [Release Notes](#).

For network controllers, the **Port Information** card displays a visual overview of the ports of the selected network controller. Additional information for some of the ports is available under **Port Details**.

Console

Note: All or some of this functionality requires site version 26.20 or later. For more information, see the document [Release Notes](#).

For network controllers, the **Console** card provides access to a Command Line Interface (CLI) in a console for troubleshooting.

A new session can be started by clicking **Open Console**. During a session, the log for the current session can be downloaded by clicking **Download Console Logs**. Logs for previous sessions are not available for download. The session can be closed manually by clicking **Close Console**. Ongoing sessions close automatically after 5 minutes of inactivity.

For a list of available commands and subcommands, see [Console Commands](#) on page 64. For more detailed descriptions and additional examples, use the `help` command in the console.

The CLI is used during the standard troubleshooting process. For more information, see [Troubleshooting Guideline](#) and [Advanced Troubleshooting Guideline](#).

6.1.2

Edit Equipment

For information about the user role access rights for this page, see the section *Local Network Management Portal Access Rights* in the document [User Role Description](#).

Note: All or some of this functionality requires site version 26.20 or later. For more information, see the document [Release Notes](#).

The configuration properties described in the following sections can be edited for the listed equipment types by clicking on the pen icons next to the properties:

Equipment Name

- Indoor radio units
- Radio dots
- Micro radios



- Macro radios
- Antenna integrated radios

TDD Pattern and Sub-pattern

The TDD pattern or sub-pattern can be edited at any time for radios. For more information, see [TDD Patterns and Sub-Patterns](#) on page 75 and [Options, Requirements, and Limitations](#) on page 80.

Note: The TDD pattern or sub-pattern cannot be configured for locations using FDD bands.

Configuring the TDD pattern or sub-pattern after the network controllers have been activated on a site will cause a short network interruption, affecting all reconfigured equipment.

Transmit Power

- Indoor radio units
- Micro radios
- Macro radios
- Antenna integrated radios



Caution!

Excessive RF exposure might result in potentially adverse health effects. If it is suspected that RF exposure limits might be exceeded, ensure that transmitting antennas are switched off, or reduce output power whilst working with or near antennas.



- Note:**
- The transmit power should generally only be changed as part of the Ericsson Private 5G installation process, according to the radio deployment plan for the site. If the site is experiencing issues with coverage and reception, contact support before making changes.
 - To change the transmit power of unknown radios, contact support.
 - The transmit power for an IRU affects all radio dots that are connected to it.
 - The transmit power for an IRU pair will be applied for both IRUs as it affects all radio dots connected to the pair.
 - For CBRS sites using SAS, the maximum allowed transmit power can be configured for each IRU, micro radio, macro radio, and antenna integrated radio. The current transmit power will never exceed the value entered for the maximum allowed transmit power, even if it is lower than the momentary maximum value allowed by SAS.

Beamforming

Antenna Integrated Radios allow for customization of the radiation pattern through beamforming. The available beamforming configuration properties are **Digital Tilt** and **Cell Shape**.

Tracking Area Code

The Tracking Area Code (TAC) can be edited for each cell of a radio. By default, every cell uses the TAC configured for the baseband during the activation of the radio access network. The default TAC cannot be edited after the radio access network activation. During runtime, a TAC can be configured for each cell, to be used for that cell only. Editing the TAC for a cell may cause a brief traffic interruption for that cell.

6.2 RADIUS

For information about the user role access rights for this page, see the section *Local Network Management Portal Access Rights* in the document *User Role Description*.

The **RADIUS Profiles** page displays information about added Remote Authentication Dial-In User Service (RADIUS) profiles. The main purpose of this is to provide a possibility to handle secondary RADIUS based user authentication and accounting from a remote server. For more information, see *Security User Guide*.



6.2.1 Add RADIUS Profile

Adding RADIUS profiles is necessary for the functionality to work. Each RADIUS profile must be assigned to a segment. Multiple segments can be assigned to a single profile. Only one profile can be assigned to a segment.

Prerequisites

The RADIUS server is configured and online.

Note: The RADIUS server configuration is the responsibility of the customer.

Steps

1. Click + **Add RADIUS Profile**.
2. Enter a name in the **RADIUS Profile Name** field.
3. Select the assigned segment from the **VRF Binding** drop-down list.
4. Click + **Add Configuration** to set up the **Authentication Server Information**.

The settings must match the setting of the configured RADIUS authentication server.

- a. Select **5G Authentication Method**.

- **Legacy User Information** must be used for 4G networks.

For 5G, the user name and password from the EPCO will be used. For 4G, the EPCO field is always used if available.

- **EAP Message** can be used for 5G networks.

EAP authentication is only applicable for 5G and if enabled, it will be used for all 5G sessions. 4G session will use PAP/CHAP even if EAP is enabled. When EAP is enabled, **EAP Secret** is required.

The EAP configuration only applies to the secondary based RADIUS configuration.

Note: EAP based secondary authentication must be supported by all participating devices.

If no authentication response is received from the primary RADIUS server by the PGW/SMF, the request is rejected unless **Assume Positive** is configured. The primary RADIUS server will be quarantined and if multiple servers are configured, a new primary RADIUS server may be selected.

- b. Click the checkbox to enable or disable **Multicast**.



The parameter configures whether RADIUS messages are sent to a single RADIUS server or multicast to several RADIUS servers.

- c. Click the checkbox to enable or disable **Disable CHAP Challenge Padding**.
- d. Click the checkbox to enable or disable **Assume Positive**.

Note: The **Assume Positive** checkbox is not available if **EAP Message** is selected as **5G Authentication Method**.

Assume Positive sets what the result is when the RADIUS server does not provide an authentication accept response.

If enabled, in cases where there is no accept response from the RADIUS server, the authentication will be handled as if valid from the side of the RADIUS server. Sessions created this way have a timeout limit.

If RADIUS servers are down because the RADIUS authentication servers are not responding, the PGW/SMF also assumes a positive authentication result.

- e. Click the checkbox to enable or disable **Override User Information**.

Enabling **Override User Information** allows for setting a specific authentication username for this RADIUS profile.

- f. If **Override User Information** is enabled, select an authentication username from the **Authentication Username** drop-down list.
- g. Enter the server IP addresses in the **Authentication Server IP Address** field.

The server IP address must be an unallocated IP address. This IP cannot be part of the target VRF segment.

There can be multiple server IP addresses in a RADIUS profile, for redundancy purposes. At least two servers are recommended.

- h. If **Multicast** is enabled, enter the number of multicast servers in the **Multicast Servers** field.

To enable multicast for a profile, a number of multicast servers must be specified. The amount must be less than the amount of servers configured. The multicast servers are selected in chronological order from the servers listed in the **Authentication Server IP Address** field. The first non-multicast server in the list will be designated as primary.

The PGW/SMF will send a copy of each message to all multicast servers, after receiving a response from the primary RADIUS server. The PGW/SMF does not wait for multicast RADIUS servers to



respond and discards any responses received from multicast servers. Each message sent to a multicast server includes the server's secret.

- i. Enter the server port in the **Port** field. If there are multiple servers, the port must be the same.

- j. Enter the client IP address in the **Client IP** field.

The client IP address is the address the network controller uses to communicate with the RADIUS server.

- k. Enter the timeout limit, measured in seconds, in the **Timeout** field.

- l. Enter the number of retry attempts in the **Retry** field.

The parameter represents the total number of attempts to send the RADIUS message before the RADIUS server is put in quarantine. The actual number of retries will be one less than the configured value.

- m. Enter the timeout limit, measured in seconds, in the **Server Down Timeout** field.

Configuring the **Server Down Timeout** is optional. If configured, RADIUS servers detected as offline, will be put in quarantine. The **Server Down Timeout** specifies the duration of the quarantine period. No RADIUS messages will be sent to a server while it is in quarantine.

When sending a RADIUS message, the PGW/SMF will wait the configured number of seconds in **Timeout** for a response from the RADIUS server. If the PGW/SMF does not receive a response within the time frame, it will retry the RADIUS message. The number of retry attempts is determined by the **Retry** value. If no response is received within $\text{Timeout} * \text{Retry}$ seconds, the RADIUS server will be put in quarantine for as long as configured in the **Server Down Timeout** field.

If multiple servers are configured, the PGW/SMF will select a new primary RADIUS server from the list of remaining non-quarantined servers. The new primary RADIUS server will be used to send the next RADIUS messages.

Note: If all servers in the list are put in quarantine, the PGW/SMF will not select a new primary RADIUS server, and no attempts to send any RADIUS messages will be made until the **Server Down Timeout** has expired for the newly-quarantined RADIUS server.

- n. Enter the secret associated with the RADIUS server in the **Secret** field.
- o. If **EAP Message** is enabled, enter the EAP secret in the **EAP Secret** field.



- p. Click **Delete Configuration** to delete the authentication server configuration.
- 5. Click + **Add Configuration** to set up the **Accounting Server Information**.
 - a. Click the checkbox to enable or disable **Multicast**.

The parameter configures whether RADIUS messages are sent to a single RADIUS server or multicast to several RADIUS servers.
 - b. Click the checkbox to enable or disable **Force Multicast**.

If forced RADIUS multicast is enabled, the PGW/SMF sends copies of accounting messages to all multicast RADIUS accounting servers without waiting for a response from the active RADIUS server.
 - c. Click the checkbox to enable or disable **Accounting On**.

If enabled, PGW/SMF will send accounting on messages to the RADIUS server. There will be no accounting start requests sent from PGW/SMF until a response is received for the accounting on message.
 - d. If **Accounting On** is enabled, click the **Accounting On Response Needed** to match the server settings.

If **Accounting On Response Needed** is disabled, an accounting on response is not required but the accounting on messages will still be sent.
 - e. If **Accounting On Response Needed** is disabled, click the **Accounting On Retry** checkbox to match the server settings.

If **Accounting On Retry** is enabled, accounting on messages will be sent until a successful reply is received.
 - f. Click the checkbox to enable or disable **Unacknowledged Mode**.

If enabled, accounting start and stop responses are ignored.
 - g. Click the checkbox to enable or disable **Interim Updates**.

If enabling or disabling **Interim Updates**, or changing **Interim Update Interval**, all PDU sessions connected to APNs using the RADIUS profile will be disconnected and have to reconnect.
 - h. Enter the interim update interval, measured in seconds, in the **Interim Update Interval** field.
 - i. Enter the server IP addresses in the **Accounting Server IP Address** field.

The server IP address must be an unallocated IP address. This IP cannot be part of the target VRF segment.



There can be multiple server IP addresses in a RADIUS profile, for redundancy purposes. At least two servers are recommended.

- j. If **Multicast** is enabled, enter the number of multicast servers in the **Multicast Servers** field.

To enable multicast for a profile, a number of multicast servers must be specified. The amount must be less than the amount of servers configured. The multicast servers are selected in chronological order from the servers listed in the **Accounting Server IP Address** field. The first non-multicast server in the list will be designated as primary.

The PGW/SMF will send a copy of each message to all multicast servers, after receiving a response from the primary RADIUS server. The PGW/SMF does not wait for multicast RADIUS servers to respond and discards any responses received from multicast servers. Each message sent to a multicast server includes the server's secret.

- k. Enter the server port in the **Port** field. If there are multiple servers, the port must be the same.
- l. Enter the client IP address in the **Client IP Address** field.

The client IP address is the address the network controller uses to communicate with the RADIUS server.

- m. Enter the timeout limit, measured in seconds, in the **Timeout** field.
- n. Enter the number of retry attempts in the **Retry** field.

The parameter represents the total number of attempts to send the RADIUS message before the RADIUS server is put in quarantine. The actual number of retries will be one less than the configured value.

- o. Enter the timeout limit, measured in seconds, in the **Server Down Timeout** field.

Configuring the **Server Down Timeout** is optional. If configured, RADIUS servers detected as offline, will be put in quarantine. The **Server Down Timeout** specifies the duration of the quarantine period. No RADIUS messages will be sent to a server while it is in quarantine.

When sending a RADIUS message, the PGW/SMF will wait the configured number of seconds in **Timeout** for a response from the RADIUS server. If the PGW/SMF does not receive a response within the time frame, it will retry the RADIUS message. The number of retry attempts is determined by the **Retry** value. If no response is received within $\text{Timeout} * \text{Retry}$ seconds, the RADIUS server will be put in quarantine for as long as configured in the **Server Down Timeout** field.



If multiple servers are configured, the PGW/SMF will select a new primary RADIUS server from the list of remaining non-quarantined servers. The new primary RADIUS server will be used to send the next RADIUS messages.

Note: If all servers in the list are put in quarantine, the PGW/SMF will not select a new primary RADIUS server, and no attempts to send any RADIUS messages will be made until the **Server Down Timeout** has expired for the newly-quarantined RADIUS server.

- p. Enter the secret associated with the RADIUS server in the **Secret** field.
- q. Click **Delete Configuration** to delete the accounting server configuration.

6. Click **Save**.

6.2.2 Edit RADIUS Profile

Editing RADIUS profiles is possible on the overview page, which lists the number of existing RADIUS profiles. Each RADIUS profile must be assigned to a segment. Multiple profiles can be assigned to a single segment. Only one segment can be assigned to a profile.

Prerequisites

The RADIUS server is configured and online.

Note: The RADIUS server configuration is the responsibility of the customer. The configuration is not validated when saved.

Steps

1. Click on the action menu  to the right of a RADIUS Profile and select **View and Edit**.
2. Click on the pen icons to edit any information.
3. Click  to confirm edits or  to discard edits.

Note: If enabling or disabling the **Interim Updates**, or changing the **Interim Update Interval**, all PDU sessions connected to APNs using the RADIUS profile will be disconnected and have to reconnect.

4. Ensure that all adjustments are accurate and compatible with the configurations of associated SIM cards to prevent any disruption.



Note: Incorrect configuration of RADIUS profile servers can affect connectivity for SIM cards utilizing the RADIUS profile.

5. Click **Save**.
6. Click **Confirm** to confirm the edits.

6.2.3 Delete RADIUS Profile

Note: A RADIUS Profile cannot be deleted if it is attached to a segment.

Steps

1. Click on the action menu  to the right of a RADIUS Profile and select **Delete**.
2. Click **Delete RADIUS Profile** to confirm.

6.3 VRFs

For information about the user role access rights for this page, see the section *Local Network Management Portal Access Rights* in the document *User Role Description*.

Note: All or some of this functionality requires site version 25.80 or later. For more information, see the document *Release Notes*.

The **VRFs** page displays information about added VRFs.

Ericsson Private 5G uses VRFs and VLANs for traffic separation on the enterprise LAN and, if present, the enterprise transport network. Ericsson Private 5G supports binding network segments to VRFs, enabling end-to-end traffic isolation across the cellular network and the enterprise LAN. Ericsson Private 5G automatically creates and maps a VLAN to each VRF, enabling both layer 2 and layer 3 traffic separation. Since VLANs cannot be configured independently from the VRF they are mapped to, in the documentation library the term VRF is used to describe functionality which applies to VRFs only, such as OSPF support, as well as the functionality of the mapped VLAN.

VRFs can be added, edited, and removed to make Ericsson Private 5G integrate with the customer's network environment as desired.

The binding works so that:

- A network segment on the cellular side of the network controllers can be bound to one VRF on the enterprise LAN, tagging the traffic with a configurable VLAN ID.
- Multiple network segments can be bound to the same VRF.



6.3.1 Add VRF

Prerequisites

To better understand how an integration of Ericsson Private 5G with an enterprise LAN works, see [VRFs, Segments, IP Addresses, and the Enterprise LAN](#) on page 61.

Steps

1. Click **+ Add VRF**.

Note: The maximum number of VRFs that can be added is 50. The **+ Add VRF** button is disabled when the maximum number of VRFs has been reached.

2. Enter a name for this VRF in the **VRF Name** field.

This name is only to help identify the VRF within NMP and is not visible anywhere else.

3. Select **Untagged (default)** or **Tagged**.

Selecting **Untagged (default)** means that traffic from any segment assigned to this VRF is not tagged, and will not be a part of a specific VRF on the customer network.

Selecting **Tagged** means that traffic from any segment assigned to this VRF is tagged with the VLAN ID entered in the **VLAN ID** field.

4. If **Tagged** was selected in the previous step, enter the VLAN ID in the **VLAN ID** fields. The same VLAN ID must be set for both network controllers.
5. Enter the MTU size for the VRF interfaces in the **MTU Size** fields.

The VRF interface MTU should be configured to be at least as much as the payload MTU and maximum as much as the radio-network equipment MTU minus 48 bytes in order to avoid fragmentation. For more information about MTU configuration, see [Payload MTU](#) on page 8.
6. Enter the IP address for each network controller in the **IP Address** fields.
7. Enter the subnet mask value for each network controller in the **Subnet Mask** fields.
8. Select **VRRP** or **OSPF**.
9. Fill in the protocol configuration details for the selected protocol.



Note: For VRFs using the VRRP protocol, the **Default Gateway** field is optional. However, when this field is not set, devices on the VRF cannot communicate with devices outside of the VRF.

VRFs which use the VRRP protocol have a default route configured in NMP, while VRFs which use the OSPF protocol do not. For OSPF VRFs, user equipment cannot reach destinations on the Enterprise LAN without manual configuration.

For more information about OSPF and BFD configuration, see the section *OSPF and BFD Configuration* in the document *Enterprise Network Preparations*.

10. Click **Save**.

6.3.2

Edit VRF

The tagging option and VRF Name cannot be edited.

VRFs which use the VRRP protocol have a default route configured in NMP, while VRFs which use the OSPF protocol do not. For OSPF VRFs, user equipment cannot reach destinations on the Enterprise LAN without manual configuration.

For more information about OSPF and BFD configuration, see the section *OSPF and BFD Configuration* in the document *Enterprise Network Preparations*.

Steps

1. Click on the action menu to the right of a VRF and select **View and Edit**.
2. Click on the pen icons to edit any information.
3. Click to confirm edits or to discard edits.
4. Click **Save**.
5. Click **Change VRF configuration**.

Note: Changing the VRF configuration causes a brief traffic disruption.

6.3.3

Delete VRF

Note: A VRF cannot be deleted if it is in use by a service or has segments assigned to it.

Steps

1. Click on the action menu to the right of a VRF.



2. Select **Delete**.
3. Click **Delete VRF**.

6.4 Segments

For information about the user role access rights for this page, see the section *Local Network Management Portal Access Rights* in the document *User Role Description*.

The **Segments** page displays various information about network segments that have previously been added. The main purpose of segments is to group device traffic within the cellular network.

Note: A default segment is created during installation and will be the default choice when SIM card entries are added to the site.

A maximum number of 100 segments are supported.

6.4.1 Segments and IP Address Ranges

When adding a SIM card for a cellular device to an active Ericsson Private 5G site, the SIM card is assigned an IP address for each segment that the SIM card is attached to. The IP-address field of each segment contains a prepopulated IP address, belonging to the IP-address range associated with the respective segment. A different IP address can be entered, as long as it is part of the segment address range.

During installation of Ericsson Private 5G, a default segment is always created. Unless another range is requested by the enterprise IT administrator and entered by the Installer during installation, the prepopulated IP-address range 172.16.0.0/16 will be used. The /16 at the end indicates that the first 16 bits (representing 172.16) are used by the wider network, leaving the last 16 bits (representing 0.0) to define the IP address range which then includes 65 536 IP addresses ($256 \times 256 = 65\,536$). The first and the last addresses of the range (172.16.0.0 and 172.16.255.255) are reserved as the network address and the broadcast address, leaving a usable range from 172.16.0.1 to 172.16.255.254 available for devices.

When additional segments are added later, Ericsson Private 5G uses prepopulated ranges in the 172.16.0.0/12 address range (spanning 172.16.0.0–172.31.255.255) unless other ranges are entered. The first additional segment gets a prepopulated value of 172.17.0.0/16, the next one gets 172.18.0.0/16, and so on. The prepopulated values can always be replaced, giving the user control over how IP-address ranges are designated for segments in Ericsson Private 5G sites. Each segment within a site must have a separate IP-address range.

Avoiding addressing conflicts between segments in Ericsson Private 5G and the enterprise LAN is the responsibility of the person acting as site admin,



normally the enterprise IT administrator. It is recommended to follow industry best practices with regards to the use of public IP addresses.

Ericsson Private 5G uses some IP-address ranges as part of general system functionality. To avoid potential addressing conflicts, it is highly recommended to not use the address ranges listed in [Table 1](#) for other purposes.

Table 1 Internally-Used IP-Address Ranges to Avoid

IP Address Range	Service	Used For
10.10.200.0/24	Internal network controller services	Network controller internal k3s and VPP communication
10.10.201.0/24	Internal network controller services	Network controller internal k3s and VPP communication
169.254.0.0/16	Internal network controller services	LMT port and network controller container-internal
198.18.0.0/15	Watchdog speed test and internal network controller services	Watchdog speed test and other internal connectivity

6.4.2

About Segment-Traffic Priority

Setting the priority level of a segment implies assigning Quality of Service Class Identifiers (QCI) or 5G Quality of Service Identifiers (5QI) on traffic within that segment. These identifiers allow for the network to understand which traffic to prioritize if congestion is encountered. There are five segment-traffic priority levels to choose from when adding or editing a segment. The priority class (QCI/5QI level) of each segment-traffic priority is predefined.

For sites using the **Decoupled** radio network mode, the predefined priority classes can be edited within the range of 1-254.

Note: Priority classes within the range of 1-5 and 82-85 are defined by 3GPP and are not recommended values.

Editing predefined priority classes is not available to all user roles.

The predefined priority classes cannot be edited for sites using the **Click-to-Deploy** and **Flexible** radio network modes. [Segment Traffic Priorities](#) shows the segment-traffic priorities available in Ericsson Private 5G and their predefined priority classes, listed from lowest to highest priority.



Table 2 Segment-Traffic Priorities

Segment-Traffic Priority	Use Cases	Priority Class (QCI/5QI value)	Additional Priority Information	Radio Link Protocol (RLC) Mode
Best effort (default)	For use cases that do not require specific characteristics or segment-traffic prioritization but can be sufficiently handled with default characteristics of 4G and 5G networks.	133	<i>Best effort</i> has the lowest priority.	Acknowledged
Software download	For use cases that require high bandwidth, can handle variable bit rates, do not demand low latency, but should be prioritized over best effort, such as software and firmware downloads.	132	<i>Software download</i> has two times relative priority over <i>Best effort</i> .	Acknowledged
Voice service	For over-the-top voice service use cases that require prioritization over best effort, such as Push-to-Talk.	131	<i>Voice service</i> has two times relative priority over <i>Software download</i> and four times relative priority over <i>Best effort</i> .	Unacknowledged
Real-time video	For video use cases that require high	130	<i>Real-time video</i> has absolute priority over	Unacknowledged



Segment-Traffic Priority	Use Cases	Priority Class (QCI/5QI value)	Additional Priority Information	Radio Link Protocol (RLC) Mode
	segment-traffic priority and low packet delay, such as CCTV and real-time video on automated guided vehicles (AGV).		<i>Best effort, Software download, and Voice services, and can completely starve those segment-traffic priorities.</i>	
Real-time automation	For real-time automation use cases that require the highest segment-traffic priority and low packet delay, such as floor sensors, actuators, AGVs, and machine control using short cyclic traffic with very short datagrams. Typical cycle time from 10 ms to 100 ms and data sizes below 800 bytes. Packets that are delayed by more than 50 ms will result in packet loss.	129	<i>Real-time automation has absolute priority over Best effort, Software download, Voice services, and Real-time video, and can completely starve those segment-traffic priorities.</i>	Unacknowledged



- Note:**
- For 4G, all segment-traffic priorities use an LTE eNodeB pre-scheduling preset. For 5G, only the real-time automation segment-traffic priority uses pre-scheduling, with a buffer size of 800 bytes, a time out of 500 ms, and a Treordering setting of 20.
 - When using unacknowledged mode, the risk of packet loss is increased while the occurrence of significant latency spikes is decreased

6.4.3 DNS Servers

DNS servers on the enterprise LAN must be reachable from each respective network segment that contains devices in need of DNS services.

Configuration of the DNS setting for segments is optional.

6.4.4 Add Segment

Prerequisites

To better understand how an integration of Ericsson Private 5G with an enterprise LAN works, see [VRFs, Segments, IP Addresses, and the Enterprise LAN](#) on page 61.

Steps

1. Click **+ Add Segment**.

The maximum number of segments that can be added is 100. The **+ Add Segment** button is disabled when the maximum number of segments has been reached.

2. Enter a name in the **Segment** field.

3. **Optional:** Edit the values in the **IP Address Range** field.

For more information, see [Segments and IP Address Ranges](#) on page 28.

4. Enter a DNS server IP address in the **Primary DNS** field.

Note: DNS servers on the enterprise LAN must be reachable from each respective network segment that contains devices in need of DNS services.

5. **Optional:** Enter a second DNS server IP address in the **Secondary DNS** field.

The field can be left blank if there is no additional DNS server.

Note: The two DNS server IP addresses must be unique.



6. Select the desired segment traffic priority from the **Segment Traffic Priority** list.
7. For sites using the **Decoupled** radio network mode, align the priority classes used in the CSP-operated RAN with the priority classes of the newly selected segment traffic priority in NMP.

Note: The priority class (QCI/5QI level) of the selected segment traffic priority in NMP and the priority class used in the CSP-operated RAN must be the same. Newly selected segment traffic priorities are applied once the priority classes used in the CSP-operated RAN are aligned with the priority classes of the segment traffic priorities in NMP and the devices belonging to the edited segment have been disconnected and then reconnected to the network. This requires manually disabling and enabling the SIM cards, or powering the devices off and on. For more information about the segment traffic priorities available and the priority classes, see [About Segment-Traffic Priority](#) on page 29.

8. Select the desired VRF that traffic in this segment should be associated with from the **VRF Binding** list.

A network segment on the cellular side of the network controller can only be bound to one VRF. Multiple network segments on the cellular side of the network controller can be bound to the same VRF.

9. **Optional:** If RADIUS profiles are available, select the desired RADIUS profile to attach to the segment from the **Attach RADIUS Profile** list.
10. Tick the checkbox for allowing the segment to access Watchdog services to enable devices on the segment to connect to Watchdog services.
11. Click **Save**.
12. Disconnect all devices from the network that belong to the edited segment by manually disabling the SIM cards, or powering off the devices.

See [Disable or Enable SIM Card](#) on page 41 for information on how to disable and enable SIM cards.
13. Reconnect all devices to the network that belong to the edited segment by manually enabling the SIM cards, or powering on the devices.

6.4.5

Edit Segment

1. Click on the action menu  to the right of a Segment and select **View and Edit**.
2. Click on the pen icons to edit any information.
3. Click  to confirm edits or  to discard edits.



4. Click **Save**
5. Disconnect all devices from the network that belong to the edited segment by manually disabling the SIM cards, or powering the devices off.

Note: See [Disable or Enable SIM Card](#) on page 41 for information on how to disable and enable SIM cards.
6. Reconnect all devices to the network that belong to the edited segment by manually enabling the SIM cards, or powering the devices on.

6.4.6 Delete Segment

1. Click on the action menu  to the right of a segment.
2. Select **Delete**.

A segment cannot be deleted if any SIM cards are assigned to it.

3. Click **Delete Segment**.

6.5 SIM Cards Tab

For information about the user role access rights for this page, see the section *Local Network Management Portal Access Rights* in the document *User Role Description*.

Note: All or some of this functionality requires site version 25.80 or later. For more information, see the document *Release Notes*.

The **SIM Cards** page displays various information about all SIM cards added to the network.

- Note:**
- Only devices with Ericsson Private 5G SIM cards, provisioned specifically for a site, can connect to an Ericsson Private 5G radio network.
 - Due to the internal timers used in Ericsson Private 5G, cellular devices will be displayed as connected for up to two hours after an abrupt loss of connection. Disabling and then re-enabling the SIM card associated with the device will force an update of the segment connection.
 - The segment connection is only an indication of the current segment connection status.



6.5.1 Multiple Segments per SIM Card

SIM cards can be attached to up to 15 segments. Up to eight of the attached segments can be used at a time by applications that run on each cellular device, providing a quick way to change how segments are used on devices.

Grouping cellular devices and devices behind gateways between multiple segments enables binding towards different VRFs or IP subnets in the enterprise LAN, as well as differentiated QoS handling within Ericsson Private 5G.

Note: If more than eight segments are used by applications that run on a cellular device, the additional segments will not work and some devices can crash.

6.5.2 Routing behind Mobile Station

Using a SIM card in a gateway that supports Routing behind Mobile Station (RBMS) allows the gateway to assign IP addresses to each device behind the gateway from IP-address ranges defined in NMP. Traffic flows between non-cellular devices behind the gateway and other devices connected to the cellular network and the enterprise LAN with full visibility in both directions.

In NMP, a configuration option is selected for a SIM card, indicating that it will be used in an RBMS gateway. One or more IP-address ranges are defined for each segment that the SIM card is attached to. The gateway must then be configured with these IP-address ranges so that it can distribute IP-addresses from these ranges to the connected non-cellular devices.

The IP-address ranges must be unique and cannot be added if used by a different gateway device. The format of the IP-address ranges must be X.X.X.X/X, where the smallest prefix length (largest network) is /24 and the largest prefix length (smallest network) is /32. The following IP-address ranges cannot be used:

- 169.254.0.0/16
- 0.0.0.0/0

SIM cards used in RBMS gateways can be attached to multiple segments, allowing each device behind the gateway to be assigned a unique IP address from the segments defined in NMP.

Note: The gateway device vendor must enable the RBMS routing functionality in the gateway device. Some device vendors refer to RBMS routing functionality as framed routing.

Gateways without Support for RBMS

Gateways without support for RBMS can also be used if they are configured to use NAT. In this scenario, devices behind the gateway are assigned IP addresses



by the gateway that are not visible towards Ericsson Private 5G or the enterprise LAN.

SIM cards used in gateways without support for RBMS can be attached to multiple segments if the gateway supports it but will only be assigned one IP address per segment. The gateway configuration defines which of those IP addresses, visible towards Ericsson Private 5G, each device communicates through.

6.5.3 Add SIM Card

- Note:**
- Reuse of micro-, mini-, and nano-SIM cards that have been cut out from full-size (credit-card size) SIM cards, is not recommended.
 - Electronic SIM Cards (eSIMs) can only be provisioned through the Bulk IMSI API. For more information, see the document *API User Guide - Partner*.

Steps

1. Click **+ Add SIM Card**.
2. Enter a name in the **SIM Card Name** field.
The SIM card name must be different from any other added SIM card.
3. **Optional:** Edit the prepopulated **IMSI** field.

The IMSI number must consist of 15 digits. The first five or six digits of the IMSI number are determined by the mobile country code (MCC) and the mobile network code (MNC) that are registered with the Ericsson Private 5G site and cannot be edited.

4. Click **Next**.
5. Select one or more segments from the **Attach Segment(s)** drop-down list.

Most devices include a segment (also known as APN or DNN) setting. For a device to be able to use a segment, that segment must first be configured in NMP (see the section *Add Segment*). The SIM card must also be configured in NMP to use that segment.

If a segment is configured in the on-device settings of a device and that segment has not been configured in NMP to be used by the SIM card in the device, the default segment configured in NMP will override the on-device setting.

If a RADIUS profile is attached to one or more of the selected segments, ensure that the SIM card is compatible with the RADIUS profiles to prevent any disruption.



A SIM card can be attached to up to 15 segments (see the section *Multiple Segments per SIM Card*). For an application on a device to use a specific segment, other than the default segment, the segment must be configured both in NMP and in the application on the device.

Note: Many Android devices are not able to connect to Ericsson Private 5G sites if the on-device segment or APN or DNN setting is left blank. Therefore, it is highly recommended to enter a value. The value can but does not need to match the segment setting for the SIM card in NMP.

6. **Optional:** Edit the prepopulated **IP Address** field or fields.

The prepopulated IP address is generated from the IP address range of the chosen segment. Any manually selected IP address needs to be part of that range. If the SIM card being added requires more than eight IP addresses, see [IP-Address Assignment for SIM Cards](#) on page 44.

7. **Optional:** Select any of the available segments in the **Default Segment** drop-down list to be the default segment.

If no other segment settings have been made on the device, the device will use the default segment when connecting to the network.

8. Click **Next**.

9. If a SIM card is already inserted in the SIM card writer, continue with [Step 10](#). If a SIM card is not inserted in the SIM card writer, either click **Skip this step** to add the SIM card to the system without writing a SIM card, or insert a SIM card in the SIM card writer and continue with [Step 10](#).

A SIM card can be written at a later time. For more information, see [Edit SIM Card](#) on page 38.

10. Depending on what type of SIM card is in the SIM-card writer and how the SIM card is configured, proceed with one of the following alternatives:

- For 4G SIM cards, click **Write SIM** and continue with [Step 11](#).
- For 5G SIM cards with service 124 and service 125 on, select a protection scheme from the **Protection Scheme** drop-down list.

Depending on the selected protection scheme, also select a key ID from the **Public Key ID** drop-down list.

Note: If the **Public Key ID** drop-down list is empty, one must be added. For more information, see [Add Key](#) on page 45.

Click **Write SIM** and continue with [Step 11](#).



Note: If service 123 is off, the 5G-GUTI will be handled on the device instead of the SIM card.

If services 123, 124, and 125 are on, SUCI can be used and the 5G-GUTI will be handled on the SIM card.

- For 5G SIM cards with service 124 and service 125 off, click **Write SIM** and continue with [Step 11](#).

Note: The 5G SIM card will be written as a 4G SIM card.

11. Click + **Add another SIM card** to add more SIM cards, or click **Finish** to return to the **SIM Cards** page.

After This Task

The SIM card must be inserted into a device before the device can connect to the Ericsson Private 5G network. For information about how to insert a SIM card into a device, refer to the device manual provided by the device manufacturer.

To install an eSIM, see [Edit SIM Card](#) on page 38.

6.5.4

Edit SIM Card

Note: Reuse of micro-, mini-, and nano-SIM cards that have been cut out from full-size (credit-card size) SIM cards, is not recommended.

Steps

1. Click on the action menu  to the right of a SIM card.
2. Select **Edit**.

Click on the pen icons to edit the information described in the next steps. After edits have been made, click  to confirm or  to discard edits.

Note: Edits will only be saved after clicking on **Save** at the bottom of the page.

3. **Optional:** Enter a new name in the **SIM Card Name** field.

The SIM card name must be different from any other added SIM card.

4. **Optional:** Select **Yes** or **No** in the **SIM-to-device locking** drop-down list.

Locking a SIM card (IMSI) to a device (IMEI) assures that the SIM card or a clone of the SIM card cannot be used to grant another device access to the Ericsson Private 5G site. To use the SIM card in a different device, the SIM-to-device lock must first be disabled.



Note: This setting is not editable for SIM cards and eSIMs provisioned through the Bulk IMSI API. For SIM cards that have not been provisioned through the Bulk IMSI API, this setting becomes editable after the SIM card is inserted in a device.

5. **Optional:** Select **Yes** or **No** in the **Use SIM card in RBMS gateway** drop-down list.

- See [Routing behind Mobile Station](#) on page 35 before attending to this step and [Step 9](#).
- The gateway device vendor must enable the routing functionality (also known as RBMS or Routing Behind Mobile Station) in the gateway device for this feature to work.

Note: When selecting **No** for SIM cards in an RBMS gateway, all IP address ranges for all segments attached to the SIM card will be removed. The devices behind the gateway will lose connectivity permanently.

6. **Optional:** For eSIMs, scan the QR code or copy the activation code to install the eSIM.

Note: After successful installation, the **eSIM installed** field must be manually updated for the eSIM status to be correctly reflected in the SIM Card list.

For Apple devices, 5G SA needs to be turned on to see available networks.

7. **Optional:** Changes to attached segments:

Most devices include a segment (also known as APN or DNN) setting. For a device to be able to use a segment, that segment must first be configured in NMP (see the section *Add Segment*). The SIM card must also be configured in NMP to use that segment.

If a segment is configured in the on-device settings of a device and that segment has not been configured in NMP to be used by the SIM card in the device, the default segment configured in NMP will override the on-device setting.

If a RADIUS profile is attached to one or more of the selected segments, ensure that the SIM card is compatible with the RADIUS profiles to prevent any disruption.

A SIM card can be attached to up to 15 segments (see the section *Multiple Segments per SIM Card*). For an application on a device to use a specific segment, other than the default segment, the segment must be configured both in NMP and in the application on the device.



Note: Many Android devices are not able to connect to Ericsson Private 5G sites if the on-device segment or APN or DNN setting is left blank. Therefore, it is highly recommended to enter a value. The value can but does not need to match the segment setting for the SIM card in NMP.

- a. If the SIM card is only attached to a single segment, change it by selecting another segment from the **Segment Name** drop-down list.

The single segment serves as the default segment.

- b. If the SIM card is attached to multiple segments, choose a different default segment by selecting another segment from the **Default Segment** drop-down list.

Note: Changing the segment for a SIM card that is only attached to one segment, or changing the default segment for a SIM card attached to multiple segments, will cause the device to disconnect and then reconnect.

8. **Optional:** Enter a new IP address in the **IP Address** field for each segment.

Note: For changes to the IP addresses to take effect, manual disconnection and reconnection of the device containing the SIM card is required.

9. If **Yes** was selected in [Step 5](#), define the IP address ranges that will be routed behind this gateway device. Enter the IP address in the **IP Address Ranges** field and press space bar to confirm each IP address range.

A maximum of 20 IP address ranges can be defined for each segment. See [Routing behind Mobile Station](#) on page 35 for more information about IP address ranges that can be used.

Note: Editing the IP address ranges will cause the SIM card to disconnect temporarily. While disconnected, any device behind the gateway will lose connectivity.

10. **Optional:** Click on **+ Attach New Segment** to attach one or more segments to the SIM card.

- a. Select one or more segments from the drop-down list.
- b. Enter a new IP address in the **IP Address** field for that segment.
- c. If **Yes** was selected in [Step 5](#), define the IP address ranges that will be routed behind this gateway device. See [Step 9](#) for more information.

11. **Optional:** Click on the bin icon to detach any of the attached segments.

12. Click **Save**.



6.5.5 Disable or Enable SIM Card

Note: When disabling the SIM card for a Watchdog device and enabling it again, the user has to manually turn the device off and then on again for the Watchdog device to reconnect to the network.

Steps

1. Click on the action menu  to the right of a SIM card.
2. Select **Disable** or **Enable**.
3. Click **Disable SIM card** or **Enable SIM card**.

6.5.6 Write or Rewrite SIM Card

- Note:**
- The option to write a new SIM card is available in case a SIM card was not written when a SIM card was initially added to the system, or in case a SIM card has been lost or damaged.
 - There is no need to write a new SIM card after editing the **SIM Card Name, Segment, or IP Address** information since it is not stored on the SIM card.
 - Bulk-provisioned SIM cards cannot be rewritten.
 - Any edits mentioned in the steps below are optional when re-writing a SIM card.

Steps

1. Click on the action menu  to the right of a SIM card.
2. Select either **Write SIM card** or **Rewrite SIM card**.
3. Enter a new SIM card name.

Note: The SIM card name must be different from any other added SIM card.

4. Click **Next**.
5. Select one or more segments from the **Attach Segment(s)** drop-down list.

A SIM card can be attached to up to 15 segments. The number of available segments depends on the amount of segments that have been added to the site. For more information, see [Multiple Segments per SIM Card](#) on page 35.

6. Edit the prepopulated **IP Address** field or fields.



The prepopulated IP address is generated from the IP address range of the chosen segment. Any manually selected IP address needs to be part of that range. If the SIM card being added requires more than eight IP addresses, see [IP-Address Assignment for SIM Cards](#) on page 44.

7. **Optional:** Select any of the available segments in the **Default Segment** drop-down list to be the default segment.

If no other segment settings have been made on the device, the device will use the default segment when connecting to the network.

8. Click **Next**.
9. For sites using the **Decoupled** radio network mode, select one or more network slices from the **Attach Network Slice(s)** drop-down list or click **Proceed to the next step** to skip the network slice configurations.

Note: All or some of this functionality requires site version 25.80 or later. For more information, see the document [Release Notes](#).

Configuring specific network slicing is optional and is only applicable for devices using the 5G communication standard, proceed to the next step to skip the network slicing configurations or if the 4G communication standard is used.

A SIM card can be attached to up to eight network slices.

- a. If attaching more than one network slice, edit the **Associated Segment(s)** and the **Default Segment** for each network slice.

Click on the pen icons to edit. After edits have been made, click ✓ to confirm or ✕ to discard edits.

- b. If attaching more than one network slice, select any of the available network slices from the **Default Network Slice** drop-down list to be the default network slice.

If no other network slice settings have been made on the device, the device will use the default network slice when connecting to the network.

- c. Click **Next**.

10. If the SIM card writer is empty, follow the on-screen instructions and insert the SIM card.
11. Depending on what type of SIM card is in the SIM-card writer and how the SIM card is configured, proceed with one of the following alternatives:
 - For 4G SIM cards, click **Write SIM** and continue with [Step 12](#).



- For 5G SIM cards with service 124 and service 125 on, select a protection scheme from the **Protection Scheme** drop-down list. For Apple devices, Profile A or Profile B must be selected since they require SUCI.

Depending on the selected protection scheme, also select a key ID from the **Public Key ID** drop-down list.

Note: If the **Public Key ID** drop-down list is empty, one must be added. For more information, see [Add Key](#) on page 45.

Click **Write SIM** and continue with [Step 12](#).

Note: If service 123 is off, the 5G-GUTI will be handled on the device instead of the SIM card.

If services 123, 124, and 125 are on, SUCI can be used and the 5G-GUTI will be handled on the SIM card.

- For 5G SIM cards with service 124 and service 125 off, click **Write SIM** and continue with [Step 12](#).

Note: The 5G SIM card will be written as a 4G SIM card.

12. Click **Finish**.

After This Task

The SIM card must be inserted into a device before the device can connect to the Ericsson Private 5G network. For information about how to insert a SIM card into a device, refer to the device manual provided by the device manufacturer.

6.5.7 Delete SIM Card

1. Click on the action menu  to the right of a SIM card.
2. Select **Delete**.
3. Click **Delete SIM card**.

6.5.8 Open SIM Card Information

To see more information about a selected SIM card, click on the name of the SIM card or click  to the right of the SIM card and select **View Details**. See [SIM Card Information](#) on page 43 for more information.

6.5.9 SIM Card Information

The **SIM Card Information** page displays the following information about the selected SIM card:



- The **SIM Cards Information** card displays the name, type, IMSI, IMEI, ICCID, and SIM-to-Device lock status of the device associated with the SIM card.
- The **Segment Information** card displays all segments that have been attached to the SIM card and their connection status, segment priority, and IP address.

6.5.10 IP-Address Assignment for SIM Cards

For each segment that a SIM card is associated with, Ericsson Private 5G assigns one IPv4 address (which can be regarded as an IPv4 range with a /32 netmask). As segments are attached to SIM card entries, the IP address field is prepopulated with a suggested IP address. The prepopulated IP address can be changed by the user.

Ericsson Private 5G allows for device-to-device communication. However, some devices take the liberty to extend their own netmask, thereby disrupting device-to-device communication between themselves and other devices. A /29 netmask is commonly used by some devices.

For example, if device A is configured with the IP address 172.16.1.1 and device B is configured with the IP address 172.16.1.2, communication between device A and B will not work if the software in any of the two devices use a /29 netmask internally.

Ericsson Private 5G tries to accommodate this type of behavior by leaving seven extra IP addresses unused between the prepopulated IP address and the last created IP address in each segment.

Leaving space for the extra seven IP addresses will, in most cases, avoid potential conflicts. Some devices can allocate larger netmasks and make more than eight IP addresses unreachable. The responsibility to avoid IP-address conflicts lays with the person acting as site admin. If device-to-device communication is not a required use case for the site, consecutive IP addresses can be assigned without leaving any addresses unused.

Note: SIM card entries can be attached to more than one segment and therefore have more than one IP address. For more information, see [Multiple Segments per SIM Card](#) on page 35.

6.5.11 Device-to-Device Communication

Communication between cellular devices connected to Ericsson Private 5G is a normal use-case and works by default depending on which segment or segments the devices are assigned to and which VRF or VRFs the segment or segments are bound to.

Enabling or disabling communication between specific devices are handled through the use of segments and VRFs. Communication between devices is:



- Enabled (routed internally within the network controllers) for devices that are assigned to the same segment
- Enabled (routed internally within the network controllers) for devices that are assigned to different segments that are bound to the same VRF
- Disabled (not routed within the network controllers) for devices that are assigned to different segments that are bound to different VRFs

Note: Traffic between devices can still be routed through the enterprise LAN connection if configured by the enterprise to enable such routing.

6.6 Identity Concealment Keys Tab

For information about the user role access rights for this page, see the section *Local Network Management Portal Access Rights* in the document *User Role Description*.

The **Identity Concealment Keys** tab displays an overview of the identity concealment keys added to the Ericsson Private 5G site. The keys are represented by key IDs that consist of numbers between 0 and 255. Each combination of a key ID and an HPLMN ID must be unique.

For keys added through NMP, the HPLMN ID will always be that of the site (the MCC and MNC of the site), and therefore, a unique public key ID should be used. For keys added through the API, the HPLMN ID is specified in the API calls, and in that case, the same key ID can be used for multiple keys as long as they use different HPLMN IDs.

For example, if key ID 1 is already used for a key that was added through NMP, further keys added through NMP can not use key ID 1. This is because the combination of key ID 1 and the HPLMN ID of the site was already used.

In another example, where key ID 5 is already used for a key that was added through the API together with a specific HPLMN ID, key ID 5 can be used for another key as long as that key uses a different HPLMN ID.

6.6.1 Add Key

1. Click **+ Add Key**.
2. Enter a value in the **Public Key ID** field.
3. Click **Add** to confirm.

6.6.2 Delete Key

To delete a key, click on the action menu  to the right of the entry and select **Delete**. Click **Delete** to confirm.



- Note:**
- Keys created through the API cannot be deleted from NMP.
 - SIM cards configured to use a key that is deleted will no longer work and must be rewritten to work again.
 - Deleting keys is not available to all user roles.

6.7 User Equipment Traces Tab

For information about the user role access rights for this page, see the section *Local Network Management Portal Access Rights* in the document *User Role Description*.

The **User Equipment Traces** tab provides a way for skilled and experienced users to access detailed user equipment (device) diagnostics for troubleshooting, monitoring, and behavior analysis. User equipment traces are available for download until one week after a trace is complete.

Note: All or some of this functionality requires site version 26.20 or later. For more information, see the document *Release Notes*.

6.7.1 Start User Equipment Trace

Note: When starting a user equipment trace, collected data can contain sensitive and privacy-specific data. Handle the information in accordance with relevant national legislation.

Steps

1. Click **Start Trace**.

Up to 16 traces can run simultaneously.

2. **Optional:** Enter a trace name in the **Name** field.

- Note:**
- The trace name must be different from all previously added trace names.
 - If no trace name is entered, a default trace name will be set automatically.

3. Select a SIM card from the **SIM Card** drop-down list.
4. Select a duration in the **Auto Stop** drop-down list.

The trace will automatically stop when the selected time has expired, but it is always possible to manually stop a started trace. For more information, see [Stop User Equipment Trace](#) on page 47.

5. Click **Start Trace**.



6.7.2 Stop User Equipment Trace

To manually stop a running user equipment trace, click the action menu  to the right of the trace and select **Stop Trace**. The trace will be saved and contain all collected data up until the time it was stopped.

6.7.3 Download User Equipment Trace File

To download the file for a stopped user equipment trace, click on the action menu  to the right of the trace and select **Download**.

Note:

- A trace file can be downloaded for up to seven days.
- Trace files are saved in .pcap format.



7 Fault Management

The **Fault Management** page displays information about added local destinations.

Fault Management (FM) data is exposed through an SNMPv3 interface which is intended to be used by SNMP managers and other Network Monitoring System (NMS) applications that in turn can provide real-time and long-term automatic monitoring of Ericsson Private 5G sites.

Note: For sites operating in the Decoupled radio network mode, only FM data from the packet core is available as the radio network is externally managed by a CSP.

Ericsson MIB files for Data Interpretation in NMS

The following Ericsson MIB files are available to help interpreting incoming FM data in an NMS:

- ERICSSON-ALARM-MIB.mib
- ERICSSON-ALARM-PC-MIB.mib
- ERICSSON-ALARM-TC-MIB.mib
- ERICSSON-ALARM-XPATH-MIB.mib
- ERICSSON-TC-MIB.mib
- ERICSSON-TOP-MIB.mib

Ericsson Private 5G SNMP Alarms and Traps

In Ericsson Private 5G, Simple Network Management Protocol (SNMP) traps are a key method for delivering real-time fault notifications from different layers of the network to Operational Support Systems (OSS). SNMP trap sources include the RAN nodes, the Packet Core components, and the network controllers. All these elements send their traps to the Fault Management (FM) Gateway, a microservice running on the network controllers.

Within the network controllers, the Prometheus instance is used to monitor endpoint metrics and detect alert conditions based on predefined rules. A dedicated nc-trap-generator process polls Prometheus alert APIs to detect new or cleared alarms. When an alarm transitions to active, nc-trap-generator issues an SNMP new active alarm trap. When the alarm clears, it sends an SNMP alarm clear trap.

SNMP trap structure and formatting follow the Ericsson Common Information Model (ECIM) and the Ericsson Alarm MIB, which define the object identifiers (OIDs) and value formats. The nc-trap-generator maintains a persistent active



alarm table, ensuring accurate tracking of active alarms and aligning them with SNMP notifications.

The [Add Local Destination](#) on page 54 section describes how to configure the FM Gateway to forward these traps to a customer OSS system, enabling unified alarm visibility.

By default, three SNMP trap receivers are configured at deployment:

- The Fault Management Gateway in the same network controller as the nc-trap-generator
- The Fault Management Gateway in the peer network controller
- The Cloud-based Fault Management Gateway

This setup ensures SNMP traps are delivered reliably both locally and centrally for redundancy and comprehensive fault monitoring.

Table 3 SNMP Alarms

eriAlarmActiveMinorType base: 9865063 (ascii "EP5G" to decimal) + FR num * 1000 + sub FR index)

1.3.6.1.4.1.19 3.183.4.1.3.5. 1.4.34 (eriAlarmActiveSpecificProblem)	1.3.6.1.4.1.19 3.183.4.1.3.5. 1.6.34 (eriAlarmActiveEventType)	1.3.6.1.6.3.1.1 .4.1.0 (OID)	1.3.6.1.4.1.19 3.183.4.1.3.5. 1.3.34 (eriAlarmActiveMinorType)	1.3.6.1.4.1.19 3.183.4.1.3.5. 1.5.34 (eriAlarmActiveManagedObject)
Server temperature above normal alarm	equipmentAlarm (5)	eriAlarmCritical/ eriAlarmCleared	9867163	ManagedElement=<nc_hostname>, Chip=<chip>, Sensor=<sensor label>
Server temperature below normal alarm	equipmentAlarm (5)	eriAlarmMinor/ eriAlarmCleared	9867164	ManagedElement=<nc_hostname>, Chip=<chip>, Sensor=<sensor label>
CBRS Proxy loss connectivity to SAS alarm	processingErrorAlarm (4)	eriAlarmMajor/ eriAlarmCleared	9868163	ManagedElement=<nc_hostname>, Service=CBRS Proxy service
CBRS Proxy license grant	processingErrorAlarm (4)	eriAlarmMajor/ eriAlarmCleared	9868164	ManagedElement=<nc_hostname>, Service=CBRS Proxy service



1.3.6.1.4.1.19 3.183.4.1.3.5. 1.4.34 (eriAlarmActiveSpecificProblem)	1.3.6.1.4.1.19 3.183.4.1.3.5. 1.6.34 (eriAlarmActiveEventType)	1.3.6.1.6.3.1.1 .4.1.0 (OID)	1.3.6.1.4.1.19 3.183.4.1.3.5. 1.3.34 (eriAlarmActiveMinorType)	1.3.6.1.4.1.19 3.183.4.1.3.5. 1.5.34 (eriAlarmActiveManagedObject)
not renewed alarm		eriAlarmCleared		Service=CBRS Proxy service
CBRS Proxy service degradation alarm	qualityOfServiceAlarm(3)	eriAlarmMinor/eriAlarmCleared	9868165	ManagedElement=<nc_hostname>, Service=CBRS Proxy service
CBRS Proxy service failure alarm	processingErrorAlarm (4)	eriAlarmMajor/eriAlarmCleared	9868166	ManagedElement=<nc_hostname>, Service=CBRS Proxy service
Subscriber Management service degradation alarm	qualityOfServiceAlarm(3)	eriAlarmMinor/eriAlarmCleared	9868263	ManagedElement=<nc_hostname>, Service=Subscriber Management service
Subscriber Management service failure alarm	processingErrorAlarm (4)	eriAlarmMajor/eriAlarmCleared	9868264	ManagedElement=<nc_hostname>, Service=Subscriber Management service
VPPC/VPP service degradation alarm	qualityOfServiceAlarm(3)	eriAlarmMinor/eriAlarmCleared	9868363	ManagedElement=<nc_hostname>, Service=VPPC/VPP service
VPPC/VPP service failure alarm	processingErrorAlarm (4)	eriAlarmMajor/eriAlarmCleared	9868364	ManagedElement=<nc_hostname>, Service=VPPC/VPP service
Metric DB Prometheus service degradation alarm	qualityOfServiceAlarm(3)	eriAlarmMinor/eriAlarmCleared	9868463	ManagedElement=<nc_hostname>, Service=Prometheus service



1.3.6.1.4.1.19 3.183.4.1.3.5. 1.4.34 (eriAlarmActiveSpecificProblem)	1.3.6.1.4.1.19 3.183.4.1.3.5. 1.6.34 (eriAlarmActiveEventType)	1.3.6.1.6.3.1.1 .4.1.0 (OID)	1.3.6.1.4.1.19 3.183.4.1.3.5. 1.3.34 (eriAlarmActiveMinorType)	1.3.6.1.4.1.19 3.183.4.1.3.5. 1.5.34 (eriAlarmActiveManagedObject)
Metric DB Prometheus service failure alarm	processingErrorAlarm (4)	eriAlarmMajor/eriAlarmCleared	9868464	ManagedElement=<nc_hostname>, Service=Prometheus service
Failed to retrieve active alerts from Prometheus API alarm	processingErrorAlarm (4)	eriAlarmMajor/eriAlarmCleared	9868465	ManagedElement=<nc_hostname>, Service=Prometheus service
Metric DB Thanos service degradation alarm	qualityOfServiceAlarm(3)	eriAlarmMinor/eriAlarmCleared	9868466	ManagedElement=<nc_hostname>, Service=Thanos service
Metric DB Thanos service failure alarm	processingErrorAlarm (4)	eriAlarmMajor/eriAlarmCleared	9868467	ManagedElement=<nc_hostname>, Service=Thanos service
Server fan failure alarm	equipmentAlarm (5)	eriAlarmCritical/eriAlarmCleared	9867263	ManagedElement=<nc_hostname>, Sensor=<sensor_name>
Server fan degradation alarm	qualityOfServiceAlarm(3)	eriAlarmMajor/eriAlarmCleared	9867264	ManagedElement=<nc_hostname>, Sensor=<sensor_name>
Server PSU loss of incoming power alarm	environmentalAlarm (6)	eriAlarmCritical/eriAlarmCleared	9867363	ManagedElement=<nc_hostname>, Power_Supply=<power_supply>
Server PSU incoming power voltage drop alarm	qualityOfServiceAlarm (3)	eriAlarmMajor/eriAlarmCleared	9867364	ManagedElement=<nc_hostname>, Power_Supply=<power_supply>



1.3.6.1.4.1.19 3.183.4.1.3.5. 1.4.34 (eriAlarmActiveSpecificProblem)	1.3.6.1.4.1.19 3.183.4.1.3.5. 1.6.34 (eriAlarmActiveEventType)	1.3.6.1.6.3.1.1 4.1.0 (OID)	1.3.6.1.4.1.19 3.183.4.1.3.5. 1.3.34 (eriAlarmActiveMinorType)	1.3.6.1.4.1.19 3.183.4.1.3.5. 1.5.34 (eriAlarmActiveManagedObject)
Server PSU hardware failure	equipmentAlarm (5)	eriAlarmCritical/ eriAlarmCleared	9867365	ManagedElement=<nc_hostname>,Power_Supply=<power_supply>
Server disk failure alarm	equipmentAlarm (5)	eriAlarmCritical/ eriAlarmCleared	9867463	ManagedElement=<nc_hostname>,Disk=<disk_name>
Server disk degradation alarm	qualityOfServiceAlarm (3)	eriAlarmMajor/ eriAlarmCleared	9867464	ManagedElement=<nc_hostname>,Disk=<disk_name>
Interface link loss alarm	equipmentAlarm (5)	eriAlarmMajor/ eriAlarmCleared	9867563	ManagedElement=<nc_hostname>,Interface=<interface>
Server chassis alarm	equipmentAlarm (5)	eriAlarmMajor/ eriAlarmCleared	9867663	ManagedElement=<nc_hostname>,Sensor=<sensor_name>
VPN service degradation alarm	qualityOfServiceAlarm (3)	eriAlarmMinor/ eriAlarmCleared	9868563	ManagedElement=<nc_hostname>,Service=VPN_service
VPN service failure alarm	processingErrorAlarm (4)	eriAlarmMajor/ eriAlarmCleared	9868564	ManagedElement=<nc_hostname>,Service=VPN_service
NTP synchronization loss alarm	processingErrorAlarm (4)	eriAlarmMinor/ eriAlarmCleared	9868663	ManagedElement=<nc_hostname>,Service=NTP_synchronization



Power 6250 Alarms

Alarms from Power 6250 can be retrieved as part of the FM data by connecting it to a baseband. If a fault occurs in Power 6250, it will be detected by the connected baseband which will raise a corresponding alarm. The baseband sends the alarm to the FM gateway in the cloud management. The FM gateway then forwards the alarm to any configured SNMP trap manager.

To be able to retrieve alarms from Power 6250, an alarm cable must be installed between Power 6250 and an active baseband on the site. The alarm port on the baseband where the cable is connected must also be enabled in NMP. For more information, see the section *Connect Power 6250 Alarm Cable to Baseband* in the *Installation Guide - Click-to-Deploy*.

Faults that occur in Power 6250 can trigger the following alarms:

Alarm	Event Type	Network Element	Severity
Rectifier Comm Failure	environmentalAlarm	[Organization]_[Site]_[Baseband]	Warning
Mains Failure	environmentalAlarm	[Organization]_[Site]_[Baseband]	Major
Load Fuse Alarm	environmentalAlarm	[Organization]_[Site]_[Baseband]	Major

Ericsson Private 5G Advanced Management - Documentation Library

Additional documents with detailed information about Fault Management (FM) are available in the separate documentation library **Ericsson Private 5G - Advanced Management**. This library is available on CPI Extranet.

It is the responsibility of the Ericsson Customer Unit responsible for each partner or enterprise that will use the Fault Management feature to provide access to the **Ericsson Private 5G - Advanced Management** library through CPI Extranet.

The following sections within the library contain documents that can be relevant with regards to Fault Management:

- **Operation and Maintenance > Fault Management**
- **Operation and Maintenance > Reference Information > AMF and MME > Alarms and Alerts > Alarms**



Note: It is important to keep in mind that the documents in the **Ericsson Private 5G - Advanced Management** library contain a lot of information, some which is relevant and some which is not relevant, or sometimes even accurate, in the context of Ericsson Private 5G.

These documents were not created specifically for Ericsson Private 5G. Instead, they were created and are maintained by other Ericsson teams that are responsible for some of the individual products that are used as part of Ericsson Private 5G. These individual products can be included in custom solutions and in other products, such as Ericsson Private 5G. Because the implementation and scope differs depending on the context, so does the scope and accuracy of the documents.

Performance Management

Performance Management (PM) data is accessible through API calls. For more information, see the section *Local PM API* in the document *API User Guide - Enterprise*.

7.1 Add Local Destination

It is the responsibility of the enterprise and partner to provision and configure SNMP managers and other NMS applications to be able to receive FM data. Once an SNMP manager or other NMS application is provisioned and configured, it can be added as a destination for FM data.

Some SNMP managers and other NMS applications need the Ericsson Private 5G engine-ID to be included in their configuration. The Ericsson Private 5G engine-ID is 0x800000c101ac120004.

For local destinations, FM data is sent from both network controllers for redundancy. This results in each SNMP trap being sent twice to the SNMP manager or other NMS application.

Steps

1. Click **+ Add Local Destination**.

A maximum of five local destinations are supported.

2. Enter the required information and credentials.

SNMP V3 Destination:

- **Destination Name**
- **IP Address and Port** (for the server running the NMS application)

SNMP V3 Credentials (for the NMS application):

- **User Name**



- **Authentication Key**
- **Hash Algorithm**
 - Available choices are:
 - **SHA256**
 - **SHA384**
 - **SHA512**
- **Privacy Key**
- **Encryption Standard**

Available choices are:

- **AES192 Blumenthal**
- **AES192C Reeder**
- **AES256 Blumenthal**
- **AES256C Reeder**

3. Click **Save**.

Once a local destination has been added, heartbeats are sent out every minute. The NMS application receiving a heartbeat is confirmation that a destination has been successfully added.

7.2 Delete Local Destination

1. Click on the action menu  to the right of a local destination and select **Delete**.
2. Click **Delete Local Destination** to confirm.



8 Users

For information about the user role access rights for this page, see the section *Local Network Management Portal Access Rights* in the document *User Role Description*.

Note: All or some of this functionality requires site version 26.20 or later. For more information, see the document *Release Notes*.

The **Users** page contains a list of all user accounts created for accessing Local Network Management Portal.

Site Admin user accounts are created in NMP and cannot be edited or deleted in Local NMP. For more information, see the section *Local Access* in the document *NMP User Guide - Site Admin and Security Admin*. All other local user accounts are created in Local NMP. The following process describes the creation of a local user account:

1. Any user can create an account from the Local NMP login page.
2. Once an account is created, the email address used when creating the account must be verified.

If an email server is configured, a verification email is sent to the email address. If no email server is configured, the Site Admin or Local Admin must make sure the email address belongs to a trusted user before proceeding.

An email server can be configured on the **Edit Site** page. For more information, see [Edit Site](#) on page 6.

3. Once the email address is verified, a Site Admin or Local Admin can approve the user and assign a user role to the account. For more information, see [Approve User](#) on page 56.
4. Once the user role is assigned, the user is granted access to the applicable information in Local NMP. For more information about user roles and user role access rights in Local NMP, see the sections *Local Network Manager Portal User Roles* and *Local Network Management Portal Access Rights* in the document *User Role Description*.

8.1 Approve User

1. Click the action menu  to the right of a user and select **Approve**.

If the user email address is not verified, make sure it is a trusted user before proceeding.

2. Select a user role.



3. **Optional:** Click the **Add Security Permissions** check box to add additional security permissions to the user role.

Additional security permissions are available for all user roles except admin user roles.

4. Click **Approve User**.

8.2 Change User Role

1. Click the action menu  to the right of a user and select **Change Role**.
2. Select another user role.
3. **Optional:** Update the additional security permissions for the user role.

Additional security permissions are available for all user roles except admin user roles.

4. Click **Change User Role**.

8.3 Delete User

1. Click the action menu  to the right of a user and select **Delete**.
2. Click **Delete User**.



9 API Access

For information about the user role access rights for this page, see the section *Local Network Management Portal Access Rights* in the document *User Role Description*.

The **API Access** page displays a list of refresh tokens that are used with the local API. Refresh tokens generated by all local users are displayed. For more information on the use of refresh tokens and Ericsson Private 5G local APIs, see the document *API User Guide - Enterprise*.

Note: Refresh tokens are linked to the users that generated them. If a local user is deleted, so are any refresh tokens generated by that local user.

For this functionality to work correctly, the two network controllers must be connected and operational.

Generate Refresh Token

To generate a new refresh token, click the **Generate Token** button and enter the refresh token name and user credentials. If there are no active refresh tokens, the expiration date can also be set. For sites with site version 26.20 or later, select whether to generate the token with **Read-only** or **Read and write** permission. Confirm by clicking the **Continue** button.

The refresh token will only be displayed following generation and must be saved, since it cannot be recovered later. If a refresh token is lost, a new refresh token must be generated to replace it.

If another refresh token is generated after the first token, it will automatically inherit the validity period set for the first token. This validity period can only be changed by deleting all active refresh tokens, and generating a new first token.

Delete Refresh Token

To delete a refresh token, click on the action menu  to the right of the refresh token and select **Delete**. Confirm by pressing the **Delete Token** button. Any refresh token can be deleted, regardless on which local user generated it.



Appendix A: Time Synchronization Settings

Ericsson Private 5G has four default NTP servers. The NTP servers can be edited after the network controllers have been activated for a site and the site is online. Up to four different IP addresses can be added but only one is required.

Broadcast and multicast IP addresses are not supported.

The default NTP servers are listed below in case reversion is required:

- 162.159.200.123 (time.cloudflare.com)
- 162.159.200.1 (time.cloudflare.com)
- 216.239.35.4 (time2.google.com)
- 216.239.35.8 (time3.google.com)

- Note:**
- The default NTP servers are managed by third-party service providers and are subject to change without notice.
 - The time synchronization status for each network controller can be viewed on the **Equipment** page by clicking on the ID for each network controller in the **Network Controllers** card.
 - When a custom time synchronization configuration is used, enterprise firewalls for the cloud-management connection must still be open to the default NTP servers during initial deployment and during hardware replacement.



Appendix B: Replaced IRUs, Radio Dots, and Radios

Some radio equipment (IRUs, radio dots, and radios) that was sold with Ericsson Private 5G 1.1 is no longer offered and has been replaced with equivalent radio equipment. The equivalent radio equipment has the exact same hardware specifications, software functionality, and life cycle management as the replaced radio equipment. The replaced radio equipment is still supported, but in NMP and in some documents, it is represented by the new equivalent radio equipment.

Table 4 Replaced Radio Equipment and Equivalent Replacements

Replaced Radio Equipment	New Equivalent Radio Equipment
IRU 88Rb	IRU 8848
IRU 16Fr	IRU 1648
Dot 44Ar	Dot 4479
Dot 41Ar	Dot 4489
Dot 22Rn	Dot 2272
Dot 22Rn B38A	Dot 2272 PPDR1 B38A
Dot 21Rn	Dot 2282
Dot 21Rn B38A	Dot 2282 PPDR1 B38A
Dot 44Kr	Dot 4459
Dot 41Kr	Dot 4469
Radio 22Cu	Radio 2203
Radio 44Au	Radio 4418
Radio 44Pd	Radio 4422
Radio 22Cr	Radio 2217



Appendix C: VRFs, Segments, IP Addresses, and the Enterprise LAN

The way that Ericsson Private 5G integrates with an Enterprise LAN does not support a *flat* network where cellular devices and devices on the enterprise LAN share the same subnet. Instead, the concepts of VRFs and segments, combined with routing in the enterprise LAN are used to control the networking, defining which cellular devices and devices on the enterprise LAN can reach each other.

The following example can be helpful in understanding what an integration of Ericsson Private 5G with an enterprise LAN can look like and what the requirements on the enterprise LAN are:

Integration Example

In [Figure 1](#), two subnets are required for the connectivity between the network controllers and the layer 2 switch - one for each of two configured VRFs. The second VRF is not a requirement but used in this example to help explain the requirements and implications when using multiple VRFs. Four IP addresses are used in each VRF subnet as described in [Table 5](#). Those subnets and IP addresses are examples. The subnets and IP addresses used in a real installation are determined by the enterprise IT administrator.

On the radio-network side, segments A and B are bound to VRF 1 and segment C is bound to VRF 2. The subnets and IP addresses are examples, based on the prepopulated default values used in Ericsson Private 5G unless other values are entered.

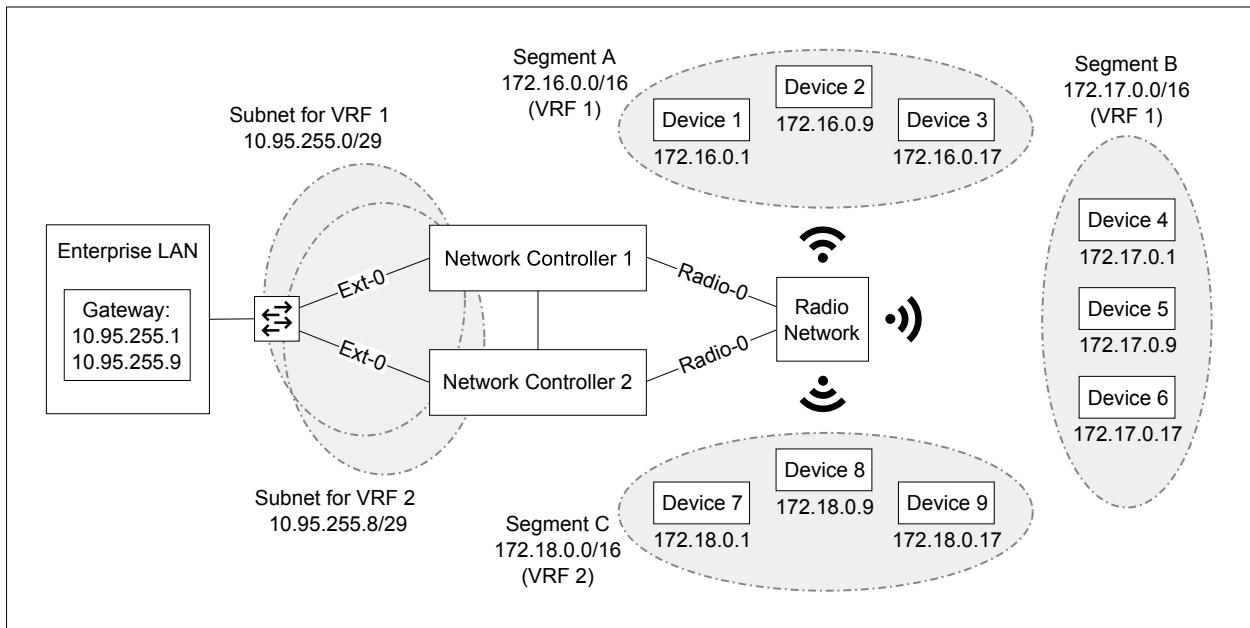


Figure 1 Example: Enterprise LAN and Cellular Device Network Integration

Table 5 VRF Subnets

VRF 1 (Ext-0), Untagged or Tagged (Default)	VRF 2 (Ext-0), Tagged
Subnet: 10.95.255.0/29	Subnet: 10.95.255.8/29
Gateway: 10.95.255.1 (in the enterprise LAN)	Gateway: 10.95.255.9 (in the enterprise LAN)
VIP: 10.95.255.2 (assigned to the presently primary network controller)	VIP: 10.95.255.10 (assigned to the presently primary network controller)
Network controller 1: 10.95.255.3	Network controller 1: 10.95.255.11
Network controller 2: 10.95.255.4	Network controller 2: 10.95.255.12

Note: To ensure redundancy of the network controllers, no other hosts are allowed in the VRF subnets.

For devices on the enterprise LAN to be able to communicate with cellular devices in segments A, B, and C, the routing detailed in [Table 6](#) must be configured on the enterprise LAN by the enterprise IT administrator.



Table 6 IP Routes Required in Enterprise LAN

Subnets Reachable on VRF 1 through VIP: 10.95.255.2	Subnets Reachable on VRF 2 through VIP: 10.95.255.10
172.16.0.0/16	172.18.0.0/16
172.17.0.0/16	



Appendix D: Console Commands

Table 7 Console Command Reference

Command	Arguments	Flags	Description
help	Option: — <COMMAND-NAME>	-h, --help	Show information and usage examples about the available commands
show firewall	Options: — table <TABLE-NAME> — One of: — netns <NETNS-NAME> — podname <POD-NAME> [namespace <NAMESPACE-NAME>]	-h, --help	Show firewall rules for subsystems running on the network controllers. For more information on firewall rules, see the section <i>VPN Internal Firewall Rules</i> in the document <i>Hardening Guideline</i>
show diameter	Subcommands: — applications Rx — applications supported-vendor-id — peer-table — transport node	-h, --help	Subcommands to show information related to Diameter — applications RX: Shows configuration and status for the Diameter Rx application. — applications supported-vendor-id: Shows supported vendor IDs for AVPs,



Command	Arguments	Flags	Description
			including non-device and application vendors. — peer-table: Shows the Diameter peer table, including configured static peers and accepted dynamic peers. — transport: Shows the Diameter transport configuration, including local endpoints used to accept or initiate connections to Diameter peers.
show gsa status		-h, --help	Shows the current operational status of the network controller.
show interface	Options: — dev <DEVICE-NAME> — vrf <VRF-NAME> — One of: — netns <NETNS-NAME> — podname <POD-NAME> [namespace]	-h, --help	Show network interfaces



Command	Arguments	Flags	Description
	<NAMESPACE-NAME>]		
show route	Options: — table <TABLE-NAME> — protocol <PROTOCOL-NAME> — vrf <VRF-NAME> — One of: — netns <NETNS-NAME> — podname <POD-NAME> [namespace <NAMESPACE-NAME>]	-h, --help	Show routing table for subsystems running on the network controllers
show topology		-h, --help	Show basebands and routers on the network
show apn	Subcommands: — statistics <APN-NAME> — user-info <APN-NAME> — apn-list	-h, --help	Subcommands to show information related to segments — statistics <APN-NAME>: Show statistics of a segment — user-info <APN-NAME>: Show information about users of a segment — apn-list: Show a list of segments



Command	Arguments	Flags	Description
show mm	Subcommands: — amf-gnodeb — diameter — mme-global-enodeb — sctp-info — user-info <IMSI> — user-info-list	-h, --help	Subcommands to show mobility management subsystem state — amf-gnodeb: Show AMF gNodeB association status — diameter: Show diameter node status — mme-global-enodeb: Show MME global eNodeB configuration — sctp-info : Show status of Stream Control Transmission Protocol (SCTP) associations and users — user-info <IMSI>: Show user info for a given IMSI — user-info-list: Show a list of user IMSIs
show node	Subcommands: — fault-management — status	-h, --help	Subcommands to show node status for software components and fault information — fault-management: Show fault



Command	Arguments	Flags	Description
			management information — status: Show node operational status information
show pgw	Subcommands: — sbi-client-status nf-type-pcf — statistics-brief — statistics-cause-code — statistics-charging — statistics-gtp — statistics-radius — statistics-shared-ip-pool — user-info <IMSI>	-h, --help	Subcommands to show PGW (Packet Gateway) statistics and state — sbi-client-status nf-type-pcf: Show status information about HTTP client connection to the PCF through SBI — statistics-cause-code: Show cause code statistics — statistics-charging: Show charging statistics — statistics-gtp: Show GPRS tunneling protocol (GTP) statistics — statistics-radius: Show RADIUS statistics — statistics-shared-ip-pool: Show shared ip pool statistics



Command	Arguments	Flags	Description
			— user-info <IMSI>: Show user information for an IMSI Note: This subcommand requires site version 26.20 or later.
show site		— --nc1 — --nc2 — --general — --ntp — -h, --help	Show site configuration and Network Time Protocol (NTP) servers. Flags are used to filter the output — --NC1: Show information for the first NC — --NC2: Show information for the second NC — --general: Show general site metadata — --ntp: Show NTP server list
show sgw	Subcommands: — interface <INTERFACE-NAME> — interface-list — statistics	— -h, --help	Subcommands to show SGW (Serving Gateway) interface and stats — interface <INTERFACE-NAME>: Show information about a specific



Command	Arguments	Flags	Description
			interface. When left empty, shows status of GTP paths — interface-list: Show list of SGW interfaces — statistics: Show SGW statistics
show link	Options: — dev <INTERFACE-NAME> — type <INTERFACE-TYPE> • vrf <VRF-NAME> — One of: — netns <NETNS-NAME> — podname <POD-NAME> [namespace <NAMESPACE-NAME>]	— -d, --details — -h, --help	Show link layer devices Note: All or some of this functionality requires site version 26.20 or later. For more information, see the document Release Notes.
ping <DESTINATION>	Options: — count <NUMBER> — size <PACKET-SIZE>	-h, --help	Test if a destination is reachable



Command	Arguments	Flags	Description
	— interface <INTERFACE-NAME> — vrf <VRF-NAME> — One of: — netns <NETNS-NAME> — podname <POD-NAME> [namespace <NAMESPACE-NAME>]		
tracpath <HOST>	Options: — print-both <true false> — packet-size <BYTES> — max-hops <MAX-HOPS> — dont-resolve <true false> — port <UDP-PORT> — vrf <VRF-NAME> — One of: — netns <NETNS-NAME> — podname <POD-NAME> [namespace	-h, --help	Show the route of a network packet from source to destination with path MTU discovery



Command	Arguments	Flags	Description
	<NAMESPACE E-NAME>]		
traceroute <HOST>	Options: — source <SOURCE-IP> — interface <INTERFACE- NAME> — max-hops <MAX-HOPS> — wait <SECONDS> — vrf <VRF- NAME> — One of: — netns <NETNS- NAME> — podname <POD- NAME> [namespace <NAMESPACE E-NAME>]	-h, --help	Show the route of a network packet from source to destination



Appendix E: Network Controller Logs

The following log types are available for the network controller logs:

EBM Logs	Event-Based Monitoring (EBM) shows real-time events related to mobility and session management, that is registration and session updates.
ISP Logs	Internet Service Provider (ISP) logs are useful for monitoring in-service performance, diagnosing crashes, and verifying the correct start up of the mobility management controller.
Domain Proxy Logs	Specific for North American CBRS deployments responsible for communication with Spectrum Allocation Server (SAS).
NTP Logs	Network Time Protocol (NTP) is used to synchronize computer clocks over a network.
VPN Client Logs	Logs relating to the secure tunnel linking the network controller back to the hosted orchestrator. This is relevant for Cloud connected deployments.
VPP Logs	Vector Packet Processing (VPP) is an user space software router that uses Data Plane Development Kit (DPDK) for efficient interface management, handles control and user plane traffic, and supports dynamic configuration and Linux synchronization via its Linux Control Plugin (LCP).
Core Network Logs	Cover multiple internal services including the following: — Vector Packet Processing Controller (VPPC) that passes configuration to VPP, sets up routes and VRFs VLAN tags. — KeyDB that syncs data such as configuration and user session information between network controllers. — Postgres, a database that is shared by several components such as the subscriber management and the speed test elements of the Watchdog. — Bird, the routing daemon that handles OSPF, BGP, and BFD protocols.



Network Function - MME/AMF Logs

Responsible for managing user equipment access and mobility, that is registration, connection management, and mobility management (handover).

- Mobility Management Entity (MME), 4G
- Access and Mobility Management Functions (AMF), 5G

Network Function - SGWC/PGWC/SMF Logs

Logs for the control plane side of the user data that is covering the following:

- Serving Gateway Control Plane (SGWC), 4G
- Packet Gateway Control Plane (PGWC), 4G
- Session Management Function (SMF), 5G

Network Function - UPF Logs

User Plane Function (UPF) is a component in the 4G/5G core network architecture that is responsible for user data processing and forwarding.

Network Function - NRF Logs

Facilitates the discovery and management of network functions. The Network Repository Function (NRF) maintains a database of available network functions within the Ericsson Private 5G network, allowing them to register, update their status, and be discovered by other network components.

Network Function - UDM/AUSF/HSS Logs

Logs covering the storage and management of user subscription and authentication data within Ericsson Private 5G networks.

- Unified Data Management (UDM), 5G
- Authentication Server Function (AUSF), 5G
- Home Subscriber Server (HSS), 4G

Appendix F: TDD Patterns and Sub-Patterns

For TDD, one single frequency range is used for both uplink and downlink communication. This is handled by dividing time into radio frames that are each 10 ms in length. Each radio frame is further divided into 20 time slots (0-19).

Each time slot can be either an uplink, a downlink, or a special slot. In Ericsson Private 5G, their configurations are referred to as TDD patterns. [Figure 2](#) shows an example of a 5G TDD pattern.

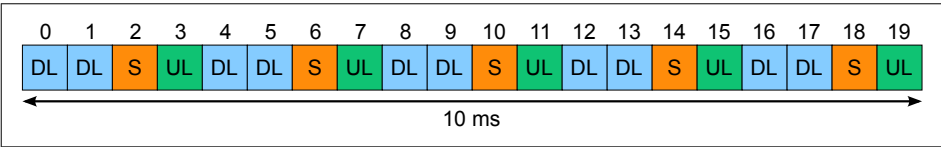


Figure 2 5G TDD Pattern 0 - DDSU

Special slots contain upload, download, and guard periods. In Ericsson Private 5G, their configurations are referred to as TDD sub-pattern. [Figure 3](#) shows an example of a TDD sub-pattern.

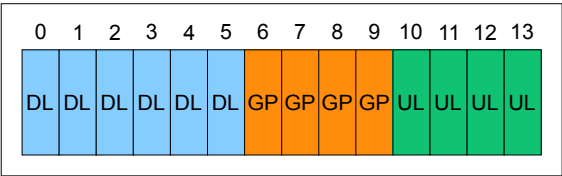


Figure 3 5G TDD Sub-Pattern 6:4:4

A radio plan should specify the TDD patterns and sub-patterns to be used for each baseband deployed with a site.

5G TDD Patterns and Sub-Patterns

[Figure 4](#)

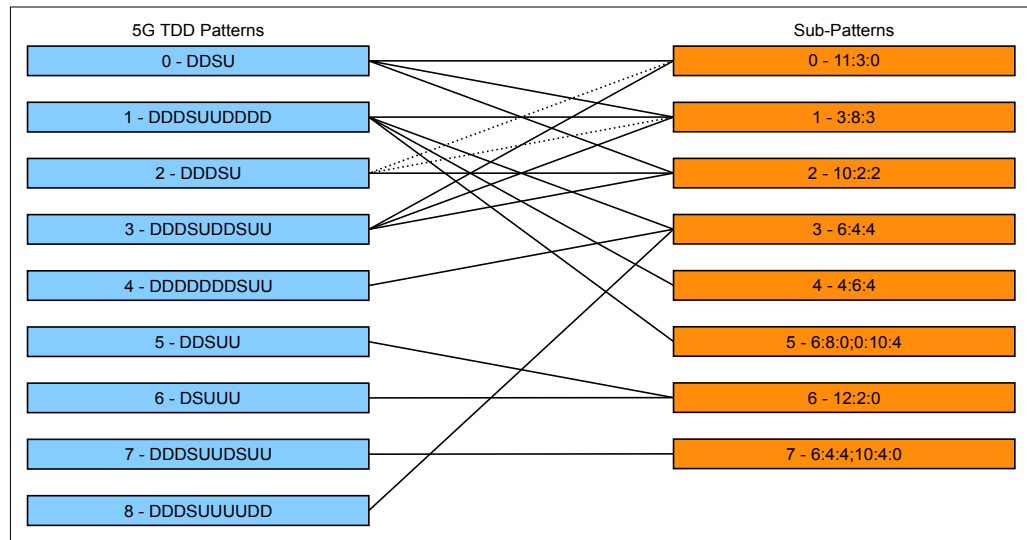


Figure 4 Combinations of 5G TDD Patterns and Sub-Patterns

- Note:**
- 5G TDD patterns 5 and 6 are only supported by radio equipment operating on the mmWave or high-band spectrum with an NR-NR Dual Connectivity configuration.
 - When using 5G TDD pattern 2 with radio equipment operating on the high-band spectrum with an NR-NR Dual Connectivity configuration, 5G TDD sub-patterns 11:3:0 and 3:8:3 are not supported. Only 10:2:2 is supported.

4G TDD Patterns and Sub-Patterns

The following 4G TDD patterns are available in Ericsson Private 5G:

- 1 - DSUUDDSUUD
- 2 - DSUDDDSUUD

Note: The default 4G TDD sub-pattern is 10:2:2 and cannot be changed.

Selecting TDD Patterns and Sub-Patterns

The following factors are usually considered when selecting TDD patterns and sub-patterns:

- Use case
- Geographic region and regulations
- Harmonization with public networks to avoid interference



Note: Allowed TDD patterns and sub-patterns vary depending on geographic region.

It is recommended to consult with support before changing TDD patterns and sub-patterns.

Uplink and Downlink Prioritization

TDD patterns were originally developed primarily for public networks that focus on downlink capacity. Most available TDD patterns favor downlink over uplink capacity, but some more than others.

To prioritize uplink capacity, TDD patterns and TDD sub-patterns with as large a proportion of uplink slots as possible should be used. To prioritize downlink capacity, TDD patterns and TDD sub-patterns with as large a proportion of downlink slots as possible should be used.

The following list includes examples of combinations of TDD patterns and sub-patterns for maximizing uplink or downlink capacity.

Maximizing uplink capacity:

- 5G TDD pattern 7 + 5G TDD sub-pattern 7 - 6:4:4;10:4:0 (45.5% uplink, 54.5% downlink)
- 5G TDD pattern 8 + 5G TDD sub-pattern 3 - 6:4:4 (44.1% uplink, 55.9% downlink)
- 5G TDD pattern 3 + 5G TDD sub-pattern 1 - 3:8:3 (39% uplink, 61% downlink)
- 5G TDD pattern 0 + 5G TDD sub-patterns 1 - 3:8:3 (35% uplink, 65% downlink)
- 4G TDD pattern 1 + 4G TDD sub-pattern 2 - 10:2:2 (44% uplink, 56% downlink)
- 5G + 4G: the available combination has 24% uplink and 76% downlink

Maximizing downlink capacity:

- 5G TDD pattern 1 + 5G TDD sub-pattern 3 - 6:4:4 (24% uplink, 76% downlink)
- 5G TDD pattern 4 + 5G TDD sub-pattern 3 - 6:4:4 (24% uplink, 76% downlink)
- 5G TDD pattern 2 + 5G TDD sub-pattern 2 - 10:2:2 (24% uplink, 76% downlink)
- 4G TDD pattern 2 + 4G TDD sub-pattern 2 - 10:2:2 (24% uplink, 76% downlink)



- 5G + 4G: the available combination has 24% uplink and 76% downlink

Note:

- The examples above do not take geographic region or harmonization with public networks into account.
- Device capability, radios, MIMO, modulation scheme, signal-to-interference ratio, and deployment conditions (elevation and blockage) can impact uplink and downlink capacity.

Harmonization with Public Networks to Avoid Interference

Harmonizing TDD patterns and sub-patterns with nearby public and private networks is usually necessary. Harmonizing is especially important if the network uses spectrum adjacent to the spectrum of a public network and is located in the coverage area of the public network. It is recommended to consult with support before changing TDD-patterns and sub-patterns.

Restrictions and Limitations

- 5G TDD pattern 7 and 8 cannot be used with Baseband 6630.
- Each individual baseband only supports a single 5G TDD pattern and 5G TDD sub-pattern or 4G TDD pattern and 4G TDD sub-pattern.
- For Click-to-Deploy sites, TDD pattern 0 and TDD sub-pattern 11:3:0 are not supported with antenna integrated radios.
- For Flexible sites, TDD pattern 0 and TDD sub-pattern 11:3:0 cannot be used together with the feature Advanced Downlink Single-User MIMO Mid-Band.
- For basebands that use CBRS and 5G, only the following TDD patterns and TDD sub-patterns can be used:
 - 5G TDD pattern 1 - DDDSUUDDDD
 - 5G TDD pattern 8 - DDDSUUUUDD
 - 5G TDD sub-pattern 3 - 6:4:4
 - 4G TDD pattern 2 - DSUDDDSUDD
 - 4G TDD sub-pattern 10:2:2
- For mixed-mode basebands that use both 4G and 5G, the compatible combinations of 4G and 5G TDD patterns are limited to a subset of the available patterns. The available combinations are:
 - 4G TDD pattern 1 - DSUDDDSUUD with 5G TDD pattern 8 - DDDSUUUUDD
 - 4G TDD pattern 2 - DSUDDDSUDD with 5G TDD pattern 1 - DDDSUUDDDD



Note: The sub-patterns for mixed-mode basebands are fixed to 4G TDD sub-pattern 10:2:2 and 5G TDD sub-pattern 6:4:4 and cannot be changed.



Appendix G: Options, Requirements, and Limitations

This chapter contains information about, requirements, options, and limitations that apply when configuring basebands. It also explains some of the concepts used within Ericsson Private 5G for baseband configuration.

Baseband Channels

Note: For site version 26.20 and later, the baseband channel concept is not valid. Instead, each cell is configured for each radio.

When used with the Click-to-Deploy radio network mode, up to six channels can be deployed on a baseband, with up to three unique bands. Channels are used within Ericsson Private 5G as a way to group radios and IRUs (and radio dots connected to the IRUs) that are connected to the same baseband and use the same radio bands. Each radio and IRU is assigned to one of these channels.

Each channel is configured to use one of three communication standards:

- **5G**
- **4G**
- **4G + 5G** (in the same channel)

Each **5G** channel and each **4G** channel is configured to use one radio band.

Each **4G + 5G** channel is configured to use one 4G band and one 5G band. Those two bands must use the same frequencies. Often, those bands have the same number, but in some cases, the 4G and 5G band numbers do not align even though they are, in essence, the "same" band as they use the same frequencies. For example, 4G band 43 uses the same frequencies as 5G band 78.



- Note:**
- With regards to the six total channels that can be deployed on a baseband, a **4G + 5G** channel counts as two. For example, if one **4G + 5G** channel is deployed, four additional **5G** or **4G** channels, or two additional **4G + 5G** channels can be deployed.
 - A maximum of three different radio bands are supported for each baseband, so in the case when more than three channels are deployed, some of those channels must be configured with the same band or bands.
 - When deploying **4G + 5G** baseband channels:
 - The two used frequency ranges cannot overlap.
 - Any gap between the two used frequency ranges cannot be larger than 10 MHz.
 - The total frequency range used, including any gap, cannot be more than 100 MHz.
 - Baseband 6631, Baseband 6630, and Radio Processor 6339 do not support deploying 5G TDD channels together with 5G FDD channels on the same baseband.

Spectrum

Spectrum is configured for each channel of each baseband for site version 25.80 and earlier and for each cell for each radio for site version 26.20 and later. Spectrum can be set either by specifying the channel bandwidth and the lower frequency border, or by specifying the channel bandwidth and an EARFCN or NR-ARFCN number.

For FDD, uplink and downlink spectrum ranges are configured separately. For TDD, there is a single shared uplink and downlink spectrum range.

Note: For sites with B48 and CBRS, see [B48 and CBRS](#) on page 82.

Additional Band Versions for Trials, Demos, and Lab-Support Activities

Additional band versions of the single band radio and radio dot types included in document [Product Package Description](#) can be configured for trials, demos, and lab support activities with Click-to-Deploy by choosing **Custom 2T2R** or **Custom 4T4R** (depending on radio or radio dot type). These band versions are not included in any Click-to-Deploy product package and have not been end-to-end verified with Ericsson Private 5G.

Note: An approved exemption request is required to use these band versions with Click-to-Deploy.

TDD Pattern

TDD pattern is a configuration option that applies only to TDD bands.



Each individual baseband only supports a single 5G TDD pattern and 5G TDD sub-pattern or 4G TDD pattern and 4G TDD sub-pattern.

If a baseband uses both 4G and 5G, it is considered a mixed-mode baseband. The compatible combinations of 4G and 5G TDD patterns and 5G TDD sub-patterns for mixed-mode basebands are currently limited to a subset of the available patterns. For more information about TDD patterns and sub-patterns, see [TDD Patterns and Sub-Patterns](#) on page 75.

Baseband Radio Package Capacity

Each baseband has a number of ports for connecting radios or the IRUs included in radio dot packages. This number is different for the different basebands. The total number of radio packages supported by a baseband depends on many factors. The following documents can help to determine the baseband radio capacity:

- Supported Radio Capabilities
- Ericsson Private 5G Supported Baseband Capacity
- CPRI Capacity

Note: The Macro 8863 packages require dual connections to the baseband and thereby use two baseband radio ports, reducing the maximum number of units that can be used.

Antenna integrated radios must be connected with a single E-CPRI connection. Use of dual CPRI connections for antenna integrated radios is not supported with Ericsson Private 5G.

B48 and CBRS

Sites with B48 and CBRS must generally use a SAS provider, a requirement of the CBRS standard. The SAS provider settings cover all basebands of a site. Sites with B48 and CBRS that operate under a Special Temporary Authorization (STA) do not use a SAS provider and instead function as any other site without B48 and CBRS.

Radio Dots

Each IRU supports eight radio dots.

Note: Some of the **Click-to-Deploy** radio dot packages contain only four radio dots but the wizard still lists **Maximum Number of Radio Dots** as eight.